

A Goal-Oriented Error Estimation Framework for Thermodynamic Topology Optimization using Adaptive Virtual Elements

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Abstract In this work, we present a fully adaptive 2D thermodynamic topology optimization framework. The numerical problem is solved using the Virtual Element Method which allows for straightforward mesh refinement. In the proposed framework, elements are marked after each optimization step based on both density change and approximation error of the strain energy, without introducing heuristic hyperparameters. The error estimation scheme utilized is a strongly simplified but highly efficient version of the goal-oriented dual-weighted residual a posteriori error estimation method.

The resulting framework produces the same optimized structures as those obtained using a fine, constant mesh. We demonstrate this property through various benchmarks, even for very fine structural features. Additionally, we introduce an adapted regularization scheme to ensure stable convergence behavior, even in the presence of large size differences among adjacent elements.

Keywords Virtual Elements · Topology Optimization · Adaptivity · Dual-Weighted-Residuals

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1 Introduction

Topology optimization is a well established tool to streamline the design process in engineering applications. The goal is to find an optimal design w.r.t. some prescribed cost function. The maximization of the material stiffness is the most common one [Sigmund and Maute \(2013\)](#) but other choices like a minimal stress design are also well established [Allaire et al. \(2004\)](#). In this work, a density based design approach with a prescribed volume fraction is utilized. Here, it is the goal to find the best material distribution. For this purpose, a continuous indicator function $\chi(\mathbf{x}) \in [\chi_{\min}, 1]$ is introduced, where 1 indicates bulk and $\chi_{\min} \ll 1$ void material. The free Helmholtz energy is then expressed as $\Psi = \chi^p \Psi_0$ where Ψ_0 is the free energy of the bulk material. To avoid nonphysical gray solutions, the material stiffness of intermediate values is further weakened by the choice of the exponent $p > 1$ [Bendsøe \(1989\)](#); [Bendsøe and Sigmund \(2004\)](#). As a consequence, the optimization problem becomes non-convex which leads to checkerboarding (void and bulk material are distributed like the black and white fields on a chess board [Díaz and Sigmund \(1995\)](#)). To overcome this problem, different filtering techniques were proposed. A good survey can be found in [Sigmund and Maute \(2013\)](#). In this work we employ the thermodynamic topology optimization (TTO), a subbranch of the standard topology optimization. Here, the structural equations as well as the update scheme for the design variable are derived from an extended Hamilton functional. With this, nonlinear material laws are easy to include. Further, no additional filtering scheme is required since the regularization is a direct part of the extended functional. The TTO roots back to [Junker and Hackl \(2015\)](#) and has been improved in [Jantos et al. \(2019a\)](#); [Junker and Balzani \(2021b\)](#) and [Błaszczuk et al. \(2022\)](#). For a comparison with the classical SIMP approach, we refer the reader to [Jantos et al. \(2019b\)](#).

To find the optimal structural design and to minimize the design space which is occupied by intermediate values of the design variable, a fine discretization of the domain is required. However, in void regions a very coarse mesh would be sufficient which motivates the coupling of topology optimization with adaptive mesh refinement (AMR). The first paper on this topic was published by [Maute and Ramm \(1995\)](#), where a fine mesh was chosen in bulk areas and a coarse one in void regions. In [Costa Jr and Alves \(2003\)](#), a marking strategy was proposed which is based on the density variable and on an energy norm error estimator. To reduce the number of elements, [Stainko \(2005\)](#) only refined elements located at structure-void interfaces. [Wang et al. \(2010\)](#) were the first who did AMR after each optimization step to reduce the dependency of the final structure on the mesh initially chosen. [Wang et al. \(2014\)](#) decoupled the density field from the analysis mesh. They use two distinct refinement indicators to further reduce the

computational cost of the refinement scheme. In [Zhang et al. \(2020b\)](#), the adaptive topology optimization was coupled with the Virtual Element Method (VEM) to compute the displacement field. The TTO was first coupled with AMR in [Vogel and Junker \(2021\)](#) and coupled with the VEM in [Cihan et al. \(2024\)](#). In this work, we also employ a mixture of marking criteria; one part of the elements is refined based on the change of the density field while the other part is refined based on an error indicator. As error estimation scheme, we employ the goal-oriented dual-weighted residual a posteriori error estimation method (DWR). This method dates back to the pioneering works of [Becker and Rannacher \(1996, 1997, 2001\)](#) and [Prudhomme and Oden \(1999\)](#); [Oden and Prudhomme \(2002\)](#). The approach has been applied in a variety of different fields since. [Richter \(2012\)](#) utilized the framework in the context of fluid-structure interaction; [Richter and Wick \(2015\)](#) for nonlinear plasticity, and [Roth et al. \(2022\)](#) for space-time formulations of nonstationary Stokes and Navier-Stokes flow. For a good survey on the topic, we refer the reader to [Endtmayer et al. \(2024\)](#)

The DWR method can be used to estimate the error for numerical solutions stemming from different numerical schemes. [Becker and Rannacher \(1997\)](#) combined the DWR scheme with the continuous Galerkin Finite Element Method, [Gonçalves et al. \(2009\)](#) with the discontinuous Galerkin Finite Element Method, [Chen and Gunzburger \(2014\)](#) with the Finite Volume Method, [Weißer and Wick \(2018\)](#) with the Boundary Element Method on polygonal meshes, [Stolfo et al. \(2019\)](#) with the Finite Cell Method, [Minakowski and Richter \(2023\)](#) with neural networks and [Sellmann and Junker \(2025\)](#) with the first order Virtual Element Method. The Virtual Element Method dates back to [Beirão Da Veiga et al. \(2013, 2014\)](#). It can be understood as a generalization of the finite element method (FEM) to arbitrarily shaped meshes. Great studies with a focus on engineering applications in structural mechanics can be found in [Artioli et al. \(2018a,b\)](#); [Wriggers et al. \(2024\)](#). A recent summary on the mathematical foundations can be found in [Beirão Da Veiga et al. \(2023\)](#). Since the VEM allows for arbitrarily shaped elements, the combination of the VEM with AMR is straightforward, which is thus the main motivation in this paper to use this method. First error estimators for the VEM were introduced in [Beirão Da Veiga and Manzini \(2015\)](#) and further investigated in [Berrone and Borio \(2017\)](#); [Beirão Da Veiga et al. \(2019\)](#); [Guo et al. \(2019\)](#); [Chi et al. \(2019\)](#); [van Huyssteen, Daniel et al. \(2022\)](#); [Du et al. \(2022\)](#).

In this paper, we extend these previous approaches by introducing three key advancements. First, we present a novel marking strategy for mesh refinement which is specifically designed for adaptive topology optimization as both the discretization error and the change in the design variable are taken into account. The framework used to estimate discretization error is a simplified version of the a posteriori goal-oriented dual-weighted residual error estimation method. [In our proposed](#)

version, we replace the analytical adjoint solution with the polynomial projection of the linear solution, avoiding an additional adjoint solve. The estimator retains local stabilization and traction contributions.

Second, we propose a novel scheme to compute the regularization term in the evolution equation of the density field χ . This scheme allows the presence of large size differences between neighboring elements without yielding oscillations in the convergence of the density field.

Lastly, we present a fully adaptive mesh refinement scheme (refinement and coarsening) utilizing the Virtual Element Method in combination with the TTO. The remainder of this paper is structured as follows. In Section 2, we review the foundations of the VEM for linear elasticity, the a posteriori goal-oriented dual-weighted residual error estimation method and of the TTO. Afterward, we present our novel marking strategy and the adapted regularization scheme in Section 3 before both are investigated numerically in Section 4. Finally, we summarize the findings of the work and give possible directions for future studies in Section 5.

2 Methods

2.1 Virtual Element Method

In this work, the VEM is used to compute the displacement from the variational formulation of the momentum equation. For the sake of completeness, we give a brief introduction to the topic in the following section. The subsequent paragraph is strongly based on the excellent introduction given in [Beirão Da Veiga et al. \(2014\)](#). A theoretical overview can be found in [Beirão Da Veiga et al. \(2023\)](#), and a monograph with a more practical focus in [Wriggers et al. \(2024\)](#).

We will first explain the VEM for scalar valued problems in 2D. Afterward, we extend the explanation to 3D where the 2D techniques are reused for each face of the elements. Finally we show the application of the VEM for the [discretization](#) of the momentum equation.

As [an](#) introductory example, we use the well known Poisson problem. Find $u : \bar{\Omega} \rightarrow \mathbb{R}$ s.t.

$$-\Delta u = f \text{ in } \Omega, \quad u = 0 \text{ on } \partial\Omega. \quad (1)$$

Here, Ω denotes the open bounded problem domain and $\partial\Omega$ its boundary with $\bar{\Omega} = \Omega \cup \partial\Omega$. With

$$\begin{aligned} H_0^1(\Omega) &:= \{v \in H^1(\Omega) : v = 0 \text{ on } \partial\Omega\} \\ A(u, \delta u) &:= (\nabla u, \nabla \delta u)_\Omega \\ F(\delta u) &:= (f, \delta u)_\Omega, \end{aligned} \quad (2)$$

the variational problem of (1) reads:

$$\text{Find } u \in V := H_0^1(\Omega) : A(u, \delta u) = F(\delta u) \quad \forall \delta u \in V. \quad (3)$$

As in the FEM, the computational domain Ω is tessellated into smaller, non-overlapping elements E . Those are either polygons in 2D or polyhedrons in 3D. The tessellation will later be referred to as $\mathcal{T}_h$ and the edges and faces of the elements as e and $\mathcal{F}$. The mesh size is $h := \max_{E \in \mathcal{T}_h} h_E$, while $h_{\min}$ denotes the smallest element diameter. Based on the decomposition, we can define the discrete variational problem based on the finite dimensional subspace $V_h \subset V$ as

$$\text{Find } u_h \in V_h : A(u_h, \delta u_h) = F(\delta u_h) \quad \forall \delta u_h \in V_h. \quad (4)$$

2.1.1 VEM in 2D

We start the explanation by introducing the VEM for 2 dimensions. To anticipate the 3D implementation, where the planar faces of polyhedra behave as 2D elements, we denote the 2D element in this section as $\mathcal{F}$ (face).

The definition of the local function space V_1 reads

$$V_1(\mathcal{F}) := \left\{ v \in C^0(\bar{\mathcal{F}}) : v|_e \in \mathbb{P}_1(e) \text{ for any edge } e \subset \partial\mathcal{F}, \Delta v = 0 \right\}. \quad (5)$$

This means that the function space consists of all C^0 functions which are first-order polynomials on the boundary ($\mathbb{P}_1$ denotes the space of all first-order polynomials) with a Laplacian equal to zero.

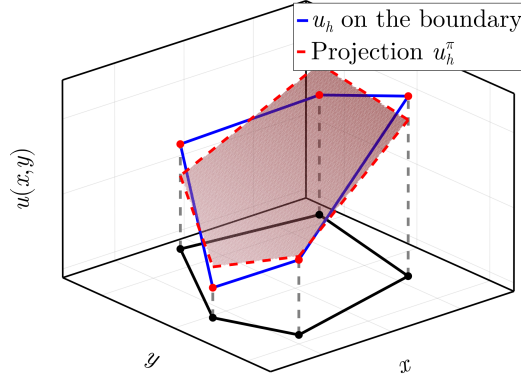
The global space V_h is a patchwork of these local spaces

$$V_h := \{v_h \in H_0^1(\Omega) : v_h|_{\mathcal{F}} \in V_1(\mathcal{F}) \text{ for all } \mathcal{F} \in \mathcal{T}_h\}. \quad (6)$$

Consequently, if we know the values of the function $u_h \in V_h$ at all nodes, u_h is also known on all edges of the mesh (inside the elements, u_h is still not known).

Clearly, knowing u_h on the boundary is not enough to solve the given boundary value problem (1). As a remedy, the linear projection operator $\Pi_{1,\mathcal{F}}^\nabla$ is introduced (the subscript $\mathcal{F}$ will be omitted in the following). It needs to be computable, solely based on known values of u_h and it should give a "good" approximation of the true unknown function. The concept of the projection can be seen in Figure 1.

As depicted, the projected function $u_h^\pi = \Pi_1^\nabla u_h$ does not coincide with u_h on the

Figure 1: Visualization of the basic idea of the projection u_h^π

element boundary. It is rather defined to fulfill the two properties

$$(\nabla(u_h - u_h^\pi), \nabla p)_\mathcal{F} = 0 \quad \forall p \in \mathbb{P}_1 \quad (7a)$$

$$\frac{1}{N^{\text{dof}}} \sum_{i=1}^{N^{\text{dof}}} (\text{dof}_i(u_h) - \text{dof}_i(u_h^\pi)) = 0. \quad (7b)$$

Here, N^{dof} is the number of nodes per element and $\text{dof}_i(u_h)$ is the function u_h evaluated at the i -th degree of freedom of the element. As shown in [Ahmad et al. \(2013\)](#), for first and second order polynomials the derived projection operator additionally fulfills the property

$$(u_h - u_h^\pi, p)_\mathcal{F} = 0 \quad \forall p \in \mathbb{P}_1. \quad (8)$$

Utilizing the introduced projection, the bilinear form can be rewritten as

$$(\nabla u_h, \nabla \delta u_h)_\mathcal{F} = \underbrace{(\nabla u_h^\pi, \nabla \delta u_h^\pi)_\mathcal{F}}_{\text{consistency}} + \underbrace{(\nabla(u_h - u_h^\pi), \nabla(\delta u_h - \delta u_h^\pi))_\mathcal{F}}_{\text{stabilization}}. \quad (9)$$

The stabilization part cannot be computed exactly, as it requires the knowledge about u_h inside the element. In the remainder of this work, we choose as approximation of the stabilization term the so called "DOFI-DOFI" stabilization (cf. [Beirão Da Veiga et al. \(2014\)](#))

$$S_\mathcal{F}(u_h, \delta u_h) := \gamma_\mathcal{F} \sum_{i=1}^{N^{\text{dof}}} \text{dof}_i(u_h - u_h^\pi) \text{dof}_i(\delta u_h - \delta u_h^\pi). \quad (10)$$

2.1.2 VEM in 3D

The definition of the VEM in 3D is quite similar to the 2D case. Only the [definition](#) of the local space $V_1(\mathcal{V})$ on the polyhedron $\mathcal{V}$ differs

$$V_1(\mathcal{V}) := \{v \in H^1(\mathcal{V}) : v|_{\mathcal{F}} \in V_1(\mathcal{F}), \Delta v = 0\}. \quad (11)$$

The projection operator $\Pi_{1,\mathcal{V}}^\nabla$ needs to fulfill the same conditions as in the two dimensional case (7). But, in order to compute the projection operator for the volume, first the projection for all the volume faces needs to be computed.

2.1.3 VEM for linear elasticity

In the following section, we explain the application of the VEM for linear elasticity. To keep things general, E denotes an element in N_D dimensions with $N_D \in \{2, 3\}$. Let $\mathbf{V} \subset [H^1(\Omega)]^{N_D}$ denote the displacement space satisfying the homogeneous Dirichlet conditions, and let $\mathbf{V}_h \subset \mathbf{V}$ be its conforming VEM subspace. The discrete variational formulation reads:

$$\text{Find } \mathbf{u}_h \in \mathbf{V}_h : (\delta \boldsymbol{\varepsilon}_h, \mathbb{C}_0 : \boldsymbol{\varepsilon}_h)_\Omega - (\mathbf{t}, \delta \mathbf{u}_h)_{\partial\Omega_N} = 0 \quad \forall \delta \mathbf{u}_h \in \mathbf{V}_h. \quad (12)$$

Here, $\boldsymbol{\varepsilon}_h := \frac{1}{2}(\nabla \mathbf{u}_h + (\nabla \mathbf{u}_h)^T)$ denotes the strain of $\mathbf{u}_h$, and $\boldsymbol{\varepsilon}_h^\pi$ denotes the strain of $\mathbf{u}_h^\pi$. The tensor $\mathbb{C}_0$ is the elastic stiffness of the fully dense material, and $\mathbf{t}$ is the traction vector on the Neumann boundary $\partial\Omega_N$.

Writing the problem formulation as sum over all elements E and utilizing the VEM projection operator $\Pi_{1,E}^\nabla$ yields

$$\begin{aligned} \sum_{E \in \mathcal{T}_h} \Big[& (\delta \boldsymbol{\varepsilon}_h^\pi, \mathbb{C}_0 : \boldsymbol{\varepsilon}_h^\pi)_E \\ & + S_E(\mathbf{u}_h, \delta \mathbf{u}_h) \\ & - (\mathbf{t}, \Pi_{1,\partial E}^\nabla \delta \mathbf{u}_h)_{\partial E \cap \partial\Omega_N} \Big] = 0 \quad \forall \delta \mathbf{u}_h \in \mathbf{V}_h. \end{aligned} \quad (13)$$

where ∂E denotes the element faces in 3D and the element edges in 2D. The stabilization parameter γ_E is chosen as $\gamma_E = \frac{1}{4}\text{tr}(\mathbb{C}_0)$ in 2D and as $\gamma_E = \frac{h_E}{4}\text{tr}(\mathbb{C}_0)$ in 3D with h_E being the element diameter.

In the following, we use the global shorthand definitions

$$\begin{aligned} A(\mathbf{v}, \mathbf{w}) &:= \sum_{E \in \mathcal{T}_h} A_E(\mathbf{v}|_E, \mathbf{w}|_E), \\ S(\mathbf{v}, \mathbf{w}) &:= \sum_{E \in \mathcal{T}_h} S_E(\mathbf{v}|_E, \mathbf{w}|_E), \\ A_h(\mathbf{v}_h, \mathbf{w}_h) &:= A(\mathbf{v}_h^\pi, \mathbf{w}_h^\pi) + S(\mathbf{v}_h, \mathbf{w}_h). \end{aligned}$$

Here, $A_E(\mathbf{v}, \mathbf{w}) := (\boldsymbol{\varepsilon}(\mathbf{v}), \mathbb{C}_0 : \boldsymbol{\varepsilon}(\mathbf{w}))_E$ is the element elasticity bilinear form, and S_E denotes the element stabilization introduced above as $S_{\mathcal{T}}$ in two dimensions, including the projection of its arguments onto the kernel of $\Pi_{1,E}^\nabla$. Projections and derivatives are evaluated elementwise, so these expressions also apply to the generally discontinuous projected fields and their remainders.

2.2 Goal-oriented error estimation

In this work we employ a simplified version of the goal-oriented dual-weighted residual error estimation (DWR) method (cf. [Becker and Rannacher \(2001\)](#); [Bangerth and Rannacher \(2003\)](#)) to estimate element-wise error contributions w.r.t. a defined quantity of interest. However, we first introduce the main ideas of the standard DWR method, as it serves later as baseline to evaluate its simplified counterpart which is introduced in Section 3.1.1. We will limit ourselves to linear partial differential equations, while the framework itself can be extended to nonlinear equations as well. For a more detailed explanation of the DWR method we refer the reader to the exhaustive review book chapter [Endtmayer et al. \(2024\)](#) and our recent work [Sellmann and Junker \(2025\)](#) where we introduced a DWR framework for the Virtual Element Method.

The aim is to estimate a quantity of interest (QoI) which is most often some technical quantity from engineering, for instance stress or displacement evaluations. This QoI is defined by a functional $J : \mathbf{V} \rightarrow \mathbb{R}$ which can have an almost arbitrary form. Examples are:

$$\bullet J(\mathbf{u}) := \int_{\Omega} \mathbf{u}[1] + \mathbf{u}[2] \, d\Omega, \quad (14a)$$

$$\bullet J(\mathbf{u}) := \frac{1}{2} \int_{\Omega} \boldsymbol{\varepsilon} : \mathbb{C}_0 : \boldsymbol{\varepsilon} \, d\Omega, \quad (14b)$$

but classical norms can be presented by $J(\mathbf{u})$ as well. Here, the notation $\mathbf{u}[1] + \mathbf{u}[2]$ refers to the sum of the first two components of $\mathbf{u}$. Using a posteriori error analysis with DWR, errors measured in $J(\mathbf{u})$ can be estimated. These can be discretization,

model or iteration errors. We denote the discrete goal functional by $J_h : \mathbf{V}_h \rightarrow \mathbb{R}$. For the strain energy in (14b), we use

$$J_h(\mathbf{v}_h) := \frac{1}{2} A_h(\mathbf{v}_h, \mathbf{v}_h).$$

For a Galerkin discretization, the error between the true and discrete goal values can be written as

$$\begin{aligned} J(\mathbf{u}) - J_h(\mathbf{u}_h) &= \frac{1}{2} \left(F(\mathbf{e}^*) - A(\mathbf{u}_h, \mathbf{e}^*) \right) \\ &\quad + \frac{1}{2} \left(D_{\mathbf{u}} J(\mathbf{u}_h)[\mathbf{e}] - A(\mathbf{e}, \mathbf{z}_h) \right) \\ &\quad + J(\mathbf{u}_h) - J_h(\mathbf{u}_h) + \mathcal{O}^3(\mathbf{e}, \mathbf{e}^*) \end{aligned} \tag{15a}$$

$$\begin{aligned} &= F(\mathbf{e}^*) - A(\mathbf{u}_h, \mathbf{e}^*) \\ &\quad + J(\mathbf{u}_h) - J_h(\mathbf{u}_h) + \mathcal{O}^2(\mathbf{e}, \mathbf{e}^*). \end{aligned} \tag{15b}$$

where we used the error $\mathbf{e} = \mathbf{u} - \mathbf{u}_h$, the adjoint error $\mathbf{e}^* = \mathbf{z} - \mathbf{z}_h$ and $D_{\mathbf{u}} J(\mathbf{u}_h)[\mathbf{e}]$ which marks the Gâteaux differential w.r.t $\mathbf{u}$. The remainder term is denoted as $\mathcal{O}$ and the superscript indicates its order of convergence with respect to the errors $\mathbf{e}$ and $\mathbf{e}^*$. The function $\mathbf{z}$ is the adjoint solution which can be interpreted as a sensitivity measure, weighting the element-wise error contributions with respect to the chosen goal functional. The adjoint solution is given by the problem:

$$\text{Find } \mathbf{z} \in \mathbf{V} : A(\delta \mathbf{u}, \mathbf{z}) = D_{\mathbf{u}} J(\mathbf{u})[\delta \mathbf{u}] \quad \forall \delta \mathbf{u} \in \mathbf{V}. \tag{16}$$

In this work, we utilize a quadratic goal functional and therefore employ the identity (15a) as the remainder term vanishes in this case.

As stated in the section before, the discrete solutions $\mathbf{z}_h$ and $\mathbf{u}_h$ are not explicitly known in the context of the VEM. We therefore approximate those by performing a triangulation of each element. For this, an additional node at the barycenter of each element is added. In consequence, each triangle consists of two adjacent boundary nodes of the element and the added barycenter node.

Remark 1. *A naive triangulation of the element using only the element nodes (without the additional node at the barycenter) yields unsatisfactory results (see Sellmann and Junker (2025)).*

The function value at the added node is calculated as

$$\mathbf{u}_0 = \mathbf{u}_0^\pi + \frac{1}{|\mathbf{E}|} \sum_{i=1}^{N^V} (\mathbf{u}_i^\pi + \mathbf{u}_{i+1}^\pi - \mathbf{u}_i - \mathbf{u}_{i+1}) |\mathbf{E}_i|, \tag{17}$$

where N^V denotes the number of nodes of the element and $|E_i|$ is the area of the i -th triangle built by the nodes $i, i+1$ and the barycenter node 0. Then, we approximate $\mathbf{u}_h$ for each triangle as a linear polynomial matching the function values at the three triangle nodes. By this we ensure that the property $\int_E \mathbf{u}_h \, dE = \int_E \mathbf{u}_h^\pi \, dE$ holds.

In Sellmann and Junker (2025) it has been shown that the stabilization error needs to be incorporated in the error identities (15) since the derivation of the DWR error estimator relies on Galerkin orthogonality between the true and approximated solution, which does not hold for the VEM approximations. For this, we introduce the bilinear correction form

$$S^{\text{corr}}(\mathbf{v}_h, \mathbf{w}_h) := A(\mathbf{v}_h - \mathbf{v}_h^\pi, \mathbf{w}_h - \mathbf{w}_h^\pi) - S(\mathbf{v}_h, \mathbf{w}_h).$$

The adapted error identities for $J(\mathbf{u}) - J_h(\mathbf{u}_h)$ read

$$\begin{aligned} J(\mathbf{u}) - J_h(\mathbf{u}_h) &= \frac{1}{2} \left(F(\mathbf{e}^*) - A(\mathbf{u}_h, \mathbf{e}^*) \right) \\ &\quad + \frac{1}{2} \left(D_{\mathbf{u}} J(\mathbf{u}_h)[\mathbf{e}] - A(\mathbf{e}, \mathbf{z}_h) \right) \\ &\quad - S^{\text{corr}}(\mathbf{u}_h, \mathbf{z}_h) + J(\mathbf{u}_h) - J_h(\mathbf{u}_h) + \mathcal{O}^3(\mathbf{e}, \mathbf{e}^*) \end{aligned} \quad (18a)$$

$$\begin{aligned} &= F(\mathbf{e}^*) - A(\mathbf{u}_h, \mathbf{e}^*) - S^{\text{corr}}(\mathbf{u}_h, \mathbf{z}_h) \\ &\quad + J(\mathbf{u}_h) - J_h(\mathbf{u}_h) + \mathcal{O}^2(\mathbf{e}, \mathbf{e}^*). \end{aligned} \quad (18b)$$

Since the functions $\mathbf{u}$ and $\mathbf{z}$ are unknown in most practical applications they need to be approximated. In this work we utilize three different techniques. Straightforward approaches are e.g. the computation of these functions on finer meshes or the utilization of higher order ansatz spaces. As baseline, we employ $\mathbf{z} \approx \mathbf{z}_h^{(2)}$ as it is done e.g. in Richter and Wick (2015); Endtmayer et al. (2024). As the computation of higher-order functions is quite expensive, different reconstruction techniques have been proposed to obtain a higher order approximation in a post-processing step. However, most of them fall short for general polygonal elements and/or if the shape functions are not known everywhere (cf. Becker and Rannacher (2001); Rognes and Logg (2012); Carpio et al. (2013)). For polygonal meshes Weißer and Wick (2018) proposed a method where each geometrical edge needs to consist of an even number of nodes. If this is not the case, additional unknowns are added to the global problem which can increase the computational cost dramatically. Therefore, we utilize the **G**auss-**P**oint **R**econstruction **M**ethod with static condensation (**GPRMSC**) as reconstruction technique which has been proposed in our work Sellmann and Junker (2025).

The third method simply uses $\mathbf{z} \approx \mathbf{z}_h^\pi$. For more details on the latter two methods

see Section 3.1.1.

2.2.1 Error localization

The error estimator presented so far does only allow for an estimation of the global error. Since we want to utilize the error estimator to indicate which elements should be refined, local error contributions are required. To do so, we utilize the partition of unity (PU) based localization strategy presented in Richter and Wick (2015). For this we use the fact that the sum of all projected shape functions always equals a vector filled with ones at every location in the domain Ω , i.e.,

$$\mathbf{1} \equiv \sum_i^m \boldsymbol{\varphi}_i^\pi(\mathbf{x}) \quad \forall \mathbf{x} \in \Omega. \quad (19)$$

Here, m denotes the total number of all shape functions. The FEM and VEM constitute by construction PUs and therefore, this idea can easily be implemented in existing codes. Afterward, we multiply the adjoint terms point-wise (denoted by $\odot$) by the projected shape functions. For the strain-energy goal, the self-adjoint identity (54) yields

$$\begin{aligned} J(\mathbf{u}) - J_h(\mathbf{u}_h) &\approx \frac{1}{2} \sum_i^m \left[F(\mathbf{e}^\star \odot \boldsymbol{\varphi}_i^\pi) - A(\mathbf{u}_h, \mathbf{e}^\star \odot \boldsymbol{\varphi}_i^\pi) \right. \\ &\quad \left. - S_i^{\text{corr}}(\mathbf{u}_h, \mathbf{z}_h) \right] = \sum_i^m \eta_i. \end{aligned} \quad (20)$$

Here, S_i^{corr} denotes the PU contribution of the correction term, with $\sum_i^m S_i^{\text{corr}} = S^{\text{corr}}$. As a result, each η_i represents the error associated with the i -th degree of freedom. The element-wise error can be calculated afterward by adding the weighted absolute error contributions of all degrees of freedom associated with the element. The weights are determined by the number of elements containing a degree of freedom. E.g. if 5 elements contain a degree of freedom its weight is $\frac{1}{5}$. In the remainder of this paper, we utilize two metrics to measure the quality of the derived error estimator. The first metric is the effectivity index, defined as

$$\text{idx}_{\text{eff}} := \left| \frac{\sum_i^m \eta_i}{J(\mathbf{u}) - J_h(\mathbf{u}_h)} \right|. \quad (21)$$

A value of one indicates that the global error is estimated perfectly. The second metric measures the quality of the error localization by summing the absolute local

error contributions, which are later used for the mesh refinement process

$$\text{id}_{\mathbf{x}_{\text{ind}}} := \frac{\sum_i^m |\eta_i|}{|J(\mathbf{u}) - J_h(\mathbf{u}_h)|}. \quad (22)$$

Again, a value of one is optimal. If the value exceeds 1, it indicates that the sum of local error contributions exceeds the global error (known as overestimation), potentially leading to the refinement of more elements than necessary.

2.3 Thermodynamic Topology Optimization

In the following section, we briefly recall the foundations of the thermodynamic topology optimization (TTO). For more details, we refer the reader to [Jantos et al. \(2019a\)](#) and [Junker and Balzani \(2021b\)](#).

The final TTO algorithm is broadly comparable to standard topology optimization approaches, but with two main differences: (1) the filter is directly integrated into the Hamilton functional which leads to a well posed problem, and (2) there is no need for an external optimization algorithm since the density update is modeled as an evolution equation over time. These features enable a straightforward integration into existing FEM software.

If strictly derived as an inverse damage model, TTO is limited to self-adjoint problems. However, the governing equations can be derived differently, broadening the space of possible applications without losing the advantages listed above.

Here, the underlying functional is closely related to the Lagrangian used to derive the equations for the DWR framework described in the previous chapter:

$$\mathcal{L}(\mathbf{u}, \mathbf{z}, \chi) = J(\mathbf{u}, \chi) - D_{\mathbf{u}}\mathcal{G}(\mathbf{u}, \chi)[\mathbf{z}] + \mathcal{D}(\chi) + \mathcal{F}(\chi) + \mathcal{C}(\chi). \quad (23)$$

The functional $\mathcal{G} := (\Psi, 1)_{\Omega} - (\mathbf{f}, \mathbf{u})_{\Omega} - (\mathbf{t}, \mathbf{u})_{\partial\Omega}$ denotes the Gibbs energy, where the Helmholtz free energy Ψ takes the form $\Psi = \frac{1}{2}(\chi^3 \nabla \mathbf{u}, \mathbb{C}_0 : \nabla \mathbf{u})$ for linear elasticity.

The variable $\chi(\mathbf{x}) \in [\chi_{\min}, 1]$ serves as a continuous design indicator, where $\chi = 1$ signifies full material and $\chi = \chi_{\min}$ represents the absence of material.

Following the SIMP (Solid Isotropic Material with Penalization) approach, the stiffness tensor is defined as

$$\mathbb{C}(\chi) = \chi^p \mathbb{C}_0,$$

where $p = 3$ is the penalization exponent used in this work. This penalization drives the solution towards either solid or void states within the design domain Ω . The functionals $\mathcal{D}$, $\mathcal{F}$, and $\mathcal{C}$ determine the behavior of the optimization algorithm.

Here,

$$\mathcal{D} := (p_{\text{diss}}^*, \dot{\chi})_{\Omega}$$

is the dissipation functional with the positive viscosity η . The dissipation force p_{diss} is derived from a dissipation functional $p_{\text{diss}} := \frac{\partial \Delta^{\text{diss}}}{\partial \dot{\chi}}$ as is common in the field of material theory. This is in accordance with the work [Junker and Balzani \(2021a\)](#). The asterisk indicates that the dissipative force is modeled and therefore the stationarity w.r.t this quantity is not required. In this work we define $\Delta^{\text{diss}} := \frac{\eta}{2} \dot{\chi}^2$. The regularization function $\mathcal{F}$ is defined as

$$\mathcal{F} := \frac{1}{2}(\beta \|\nabla \chi\|, \|\nabla \chi\|)_{\Omega},$$

where the parameter $\beta \in \mathbb{R}^+$ controls the minimum member size to ensure a checkerboard-free structure.

Finally, the constraint functional is defined as

$$\mathcal{C} = (\chi - \varrho_0, \lambda)_{\Omega} + (\gamma, \chi)_{\Omega}.$$

The first term ensures the conservation of the total structural mass throughout the optimization process via the Lagrange multiplier $\lambda \in \mathbb{R}$. The second term ensures that the design variable χ remains within the interval $[\chi_{\min}, 1]$ using the Kuhn-Tucker parameter γ , which is defined as:

$$\gamma = \begin{cases} \bar{\gamma} & : (\dot{\chi} > 0) \wedge (\chi = 1) \\ -\bar{\gamma} & : (\dot{\chi} < 0) \wedge (\chi = \chi_{\min}) \\ 0 & : \text{else} \end{cases} \quad (24)$$

Afterward, the stationary condition of $\mathcal{L}$ w.r.t $\{\mathbf{u}, \mathbf{z}, \chi\}$ yields all the governing equations required for the optimization procedure. First, the stationarity condition w.r.t $\mathbf{z}$ yields the well known balance of momentum

$$D_{\mathbf{u}}\mathcal{G}(\mathbf{u}, \chi)[\delta \mathbf{z}] = 0 \quad \forall \delta \mathbf{z} \in \mathbf{V}. \quad (25)$$

Variation w.r.t $\mathbf{u}$ yields the governing equation for the adjoint function $\mathbf{z}$

$$D_{\mathbf{u}}J(\mathbf{u}, \chi)[\delta \mathbf{u}] - D_{\mathbf{uu}}\mathcal{G}(\mathbf{u}, \chi)[\mathbf{z}, \delta \mathbf{u}] = 0 \quad \forall \delta \mathbf{u} \in \mathbf{V}. \quad (26)$$

And finally, the variation w.r.t the design variable χ yields

$$\begin{aligned} & D_{\chi}J(\mathbf{u}, \chi)[\delta \chi] - D_{\mathbf{u}, \chi}\mathcal{G}(\mathbf{u}, \chi)[\mathbf{z}, \delta \chi] + (\eta \dot{\chi}, \delta \chi)_{\Omega} \\ & + (\beta \nabla \chi, \nabla \delta \chi)_{\Omega} + (\lambda, \delta \chi)_{\Omega} + (\gamma, \delta \chi)_{\Omega} = 0 \quad \forall \delta \chi \in H^1(\Omega). \end{aligned} \quad (27)$$

In this work, we limit our scope to strain energy minimization with the goal functional $J(\mathbf{u}, \chi) = \frac{1}{2}(\chi^3 \boldsymbol{\varepsilon}, \mathbb{C}_0 : \boldsymbol{\varepsilon})$ for linear elastic materials. Under those assumptions the problem is self adjoint, and we recover the standard governing equations for the TTO as they are derived in Junker and Balzani (2021b). The equation for $\mathbf{z}$ simplifies to

$$(\chi^3 \nabla \mathbf{u}, \mathbb{C}_0 : \nabla \delta \mathbf{u}) = (\chi^3 \nabla \mathbf{z}, \mathbb{C}_0 : \nabla \delta \mathbf{u}) \quad \forall \delta \mathbf{u} \in \mathbf{V} \quad (28)$$

which directly yields $\mathbf{z} = \mathbf{u}$. However, we wish to highlight that the adjoint required for the DWR framework is the same as that required for the present optimization approach. Consequently, even for non-self-adjoint problems, the added computational cost is limited, which strongly motivates the use of the DWR method with TTO. Further, the evolution equation for χ transforms into

$$-\left(\frac{\partial \Psi}{\partial \chi}, \delta \chi\right)_\Omega + (\lambda + \gamma, \delta \chi)_\Omega + (\eta \dot{\chi}, \delta \chi)_\Omega + (\beta \nabla \chi, \nabla \delta \chi)_\Omega \ni 0 \quad \forall \delta \chi \in H^1(\Omega). \quad (29)$$

The latter is transformed into its strong form, yielding purely local update equations

$$\begin{aligned} p_\chi + \eta \dot{\chi} + \lambda + \gamma - \nabla \cdot (\beta \nabla \chi) &\ni 0 & \forall \mathbf{x} \in \Omega \\ \beta \nabla \chi \cdot \mathbf{n} &= 0 & \forall \mathbf{x} \in \partial \Omega \end{aligned} \quad (30)$$

with the thermodynamic driving force

$$p_\chi = -\frac{p}{2} \chi^{p-1} \boldsymbol{\varepsilon} : \mathbb{C}_0 : \boldsymbol{\varepsilon}. \quad (31)$$

2.3.1 Numerical treatment of the density update

The optimization is performed in a staggered manner. First, the displacement field is computed for the current density field. Afterward, the density is updated using the evolution equation (30). The update of the density variable is based on the neighbored element method (NEM), which provides a highly efficient numerical scheme. The NEM can be seen as a generalization of the finite difference method to non-regular meshes (see Jantos et al. (2019a); Junker et al. (2019); Vogel and Junker (2020); Junker and Balzani (2021b)). In the following, the algorithmic details of the scheme are explained.

First, the evolution equation (30) is discretized using an explicit Euler scheme:

$$\chi^{i+1} = \chi^i + \frac{\Delta t}{\eta} (-p_\chi^i - \lambda - \gamma + \nabla \cdot (\beta \nabla \chi^i)). \quad (32)$$

Here, i denotes the density-update step. Following [Jantos et al. \(2019a\)](#), the quantities η and β are computed from the user-defined regularization parameter β^* , which controls the minimum structure size, and the viscosity parameter η^* . Their default values are $2h_{\min}^2$ and 15, respectively.

For the numerical update, we use the positive element-averaged driving force $p_E := -|E|^{-1} \int_E p_\chi dV$. The average driving force is defined as

$$p_{\text{avg}} := \frac{\int_\Omega p_E g(\chi) dV}{\int_\Omega g(\chi) dV}, \quad g(\chi) := (\chi - \chi_{\min})(\chi_{\max} - \chi). \quad (33)$$

The viscosity and element-wise regularization parameter are then computed as

$$\eta = p_{\text{avg}} \eta^*, \quad \beta_E = \max \{ p_{\text{avg}} \beta^*, p_{\text{avg}} h_E^2 \}, \quad (34)$$

where h_E denotes the element diameter. The use of the maximum has been proposed in [Vogel and Junker \(2021\)](#) to account for large differences in element size arising from adaptive mesh refinement.

For the density update, the regularization term $\nabla \cdot (\beta \nabla \chi)$ needs to be computed. To this end, the procedure proposed in [Jantos et al. \(2019a\)](#), with the modifications suggested in [Cihan et al. \(2024\)](#), is employed and extended to account for spatial variations of β . The element-wise values of χ and β are interpreted as samples of underlying sufficiently smooth fields at the element barycentres. Their derivatives are reconstructed from neighbouring samples using local Taylor expansions. For completeness, the underlying procedure is described below. The displayed equations apply to the two-dimensional case and can be extended analogously to three dimensions.

Since the evolution equation (30) is evaluated at the element barycentres, the required derivatives are reconstructed at the same locations. A Taylor expansion of the design variable χ about the barycentre $\mathbf{x}_j$ of element j gives

$$\begin{aligned} \chi(\mathbf{x}_i) = & \chi(\mathbf{x}_j) + \frac{\partial \chi(\mathbf{x}_j)}{\partial x} \Delta x_i + \frac{\partial \chi(\mathbf{x}_j)}{\partial y} \Delta y_i \\ & + \frac{1}{2} \left[\frac{\partial^2 \chi(\mathbf{x}_j)}{\partial x^2} \Delta x_i^2 + 2 \frac{\partial^2 \chi(\mathbf{x}_j)}{\partial x \partial y} \Delta x_i \Delta y_i + \frac{\partial^2 \chi(\mathbf{x}_j)}{\partial y^2} \Delta y_i^2 \right] + \mathcal{O}(\|\mathbf{x}_i - \mathbf{x}_j\|^3). \end{aligned} \quad (35)$$

Here, $\Delta x_i = x_i - x_j$ and $\Delta y_i = y_i - y_j$. Collecting the expansions for all elements

sharing a node with element j yields the local least-squares system

$$\underbrace{\begin{bmatrix} \frac{1}{2}\Delta x_1^2 & \frac{1}{2}\Delta y_1^2 & \Delta x_1\Delta y_1 & \Delta x_1 & \Delta y_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{1}{2}\Delta x_n^2 & \frac{1}{2}\Delta y_n^2 & \Delta x_n\Delta y_n & \Delta x_n & \Delta y_n \end{bmatrix}}_{\mathbf{G}} \underbrace{\begin{bmatrix} \chi_{,xx} \\ \chi_{,yy} \\ \chi_{,xy} \\ \chi_{,x} \\ \chi_{,y} \end{bmatrix}}_{\mathbf{m}} \approx \underbrace{\begin{bmatrix} \chi_1 - \chi_j \\ \vdots \\ \chi_n - \chi_j \end{bmatrix}}_{\mathbf{b}}. \quad (36)$$

In three dimensions, the neighbourhood consists of elements sharing an edge, and the Taylor expansion includes the additional first and second derivatives involving the third coordinate.

The derivative vector $\mathbf{m}$ is approximated by the regularized least-squares solution

$$\mathbf{m} = (\mathbf{G}^T \mathbf{G} + \alpha \mathbf{I})^{-1} \mathbf{G}^T \mathbf{b}. \quad (37)$$

The term $\alpha \mathbf{I}$ stabilizes the reconstruction when the distribution of neighbouring points leads to a singular or ill-conditioned matrix $\mathbf{G}^T \mathbf{G}$. The reconstructed Laplacian and first derivatives are

$$\nabla^2 \chi_j \approx \mathbf{m}[1] + \mathbf{m}[2], \quad \chi_{,x,j} \approx \mathbf{m}[4], \quad \chi_{,y,j} \approx \mathbf{m}[5]. \quad (38)$$

For spatially varying β , the product rule gives

$$\nabla \cdot (\beta \nabla \chi) = \beta \nabla^2 \chi + \beta_{,x} \chi_{,x} + \beta_{,y} \chi_{,y}. \quad (39)$$

The derivatives of β are reconstructed using the same least-squares procedure, replacing $\chi_i - \chi_j$ by $\beta_i - \beta_j$ in the right-hand side of (36). Thus, no additional geometric reconstruction is required. The regularization term at $\mathbf{x}_j$ is approximated as

$$[\nabla \cdot (\beta \nabla \chi)]_j \approx \beta_j (\mathbf{m}[1] + \mathbf{m}[2]) + \beta_{,x,j} \mathbf{m}[4] + \beta_{,y,j} \mathbf{m}[5], \quad (40)$$

where $\beta_{,x,j}$ and $\beta_{,y,j}$ denote the reconstructed coefficient derivatives. For constant β , the additional terms vanish and the original Laplacian-based expression is recovered.

The derivative reconstructions can be assembled for all elements as sparse matrices:

$$\nabla^2 \chi \approx \mathbf{A} \chi, \quad \chi_{,x} \approx \mathbf{A}_x \chi, \quad \chi_{,y} \approx \mathbf{A}_y \chi, \quad (41)$$

with the approximations understood at the element barycentres. Consequently, the

complete regularization term can be evaluated as

$$\mathbf{r} := \mathbf{R}(\boldsymbol{\beta})\boldsymbol{\chi} \approx [\nabla \cdot (\boldsymbol{\beta}\nabla\boldsymbol{\chi})]_h, \quad (42)$$

where

$$\mathbf{R}(\boldsymbol{\beta}) = \text{diag}(\boldsymbol{\beta})\mathbf{A} + \text{diag}(\mathbf{A}_x\boldsymbol{\beta})\mathbf{A}_x + \text{diag}(\mathbf{A}_y\boldsymbol{\beta})\mathbf{A}_y. \quad (43)$$

Here, $\text{diag}(\mathbf{v})$ denotes the diagonal matrix with diagonal entries given by $\mathbf{v}$. The matrices $\mathbf{A}$, $\mathbf{A}_x$, and $\mathbf{A}_y$ depend only on the sampling geometry. They can therefore be computed once for a fixed mesh and are recomputed after each remeshing. Changes in $\boldsymbol{\beta}$ require only an update of the coefficient-dependent terms in (43), without repeating the local least-squares reconstruction.

Boundary elements

For boundary elements, the homogeneous Neumann condition $\boldsymbol{\beta}\nabla\boldsymbol{\chi} \cdot \mathbf{n} = 0$ must be enforced. Since $\boldsymbol{\beta} > 0$, this is equivalent to $\nabla\boldsymbol{\chi} \cdot \mathbf{n} = 0$. To this end, a ghost state is introduced for each boundary edge, or boundary face in three dimensions. For an element with barycentre $\mathbf{x}_j$, its position is

$$\mathbf{x}_{\text{ghost}} = \mathbf{x}_j + 2(\mathbf{p} \cdot \mathbf{n})\mathbf{n}, \quad (44)$$

where $\mathbf{n}$ is the outward unit normal and $\mathbf{p}$ connects the barycentre to an arbitrary point on the associated boundary edge or face; see Figure 2. The ghost density is set to $\chi_{\text{ghost}} = \chi_j$, representing a vanishing normal derivative. For the coefficient reconstruction, the same reflected sampling geometry is used with $\beta_{\text{ghost}} = \beta_j$.

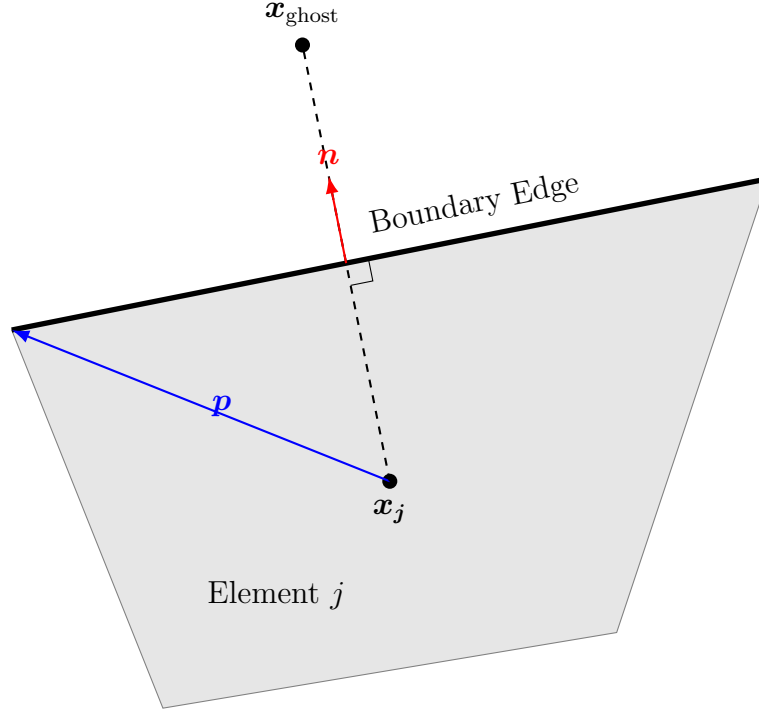


Figure 2: Computation of the ghost state position

The complete density update is summarized in Algorithm 1. The density bounds are enforced by projection onto $[\chi_{\min}, \chi_{\max}]$, while the Lagrange multiplier λ is determined by bisection to satisfy the prescribed volume fraction. The bisection interval accounts for both the driving force and the regularization contribution.

Algorithm 1 Density update.

```

1: Given:  $\chi_{\min}, \chi_{\max}, \eta^*, \beta^*, \Delta t^*, \varrho_0, h_{\min}, \epsilon_{\text{density}}$ 
2: Input:  $\mathbf{u}, \chi$ 
3: On grid change: compute  $\mathbf{A}, \mathbf{A}_x, \mathbf{A}_y$  from (36), including the ghost states
4:  $n_{\text{steps}} := \max \left\{ 1, \left\lceil \frac{12}{\eta^*} \frac{\beta^*}{h_{\min}^2} \right\rceil \right\}, \quad \Delta t := \Delta t^* / n_{\text{steps}}$ 
5: for  $i = 1, \dots, n_{\text{steps}}$  do
6:   Compute  $p_E := \frac{1}{|E|} \int_E \frac{3}{2} \chi^2 \boldsymbol{\varepsilon} : \mathbb{C}_0 : \boldsymbol{\varepsilon} \, dV$  for all  $E \in \mathcal{T}_h$ 
7:    $p_{\text{avg}} := \frac{\sum_{E \in \mathcal{T}_h} |E| p_E g(\chi_E)}{\sum_{E \in \mathcal{T}_h} |E| g(\chi_E)}, \quad g(\chi) := (\chi - \chi_{\min})(\chi_{\max} - \chi)$ 
8:    $\eta := p_{\text{avg}} \eta^*, \quad \beta_E := \max \{ p_{\text{avg}} \beta^*, p_{\text{avg}} h_E^2 \}$  for all  $E \in \mathcal{T}_h$ 
9:   Compute  $\mathbf{r} := \mathbf{R}(\boldsymbol{\beta}) \boldsymbol{\chi}$  using (43)
10:   $s_E := p_E + r_E$  for all  $E \in \mathcal{T}_h$ 
11:   $s_{\min} := \min_{E \in \mathcal{T}_h} s_E, \quad s_{\max} := \max_{E \in \mathcal{T}_h} s_E$ 
12:   $\lambda_{\min} := s_{\min} - \frac{\eta}{\Delta t} (\chi_{\max} - \chi_{\min}), \quad \lambda_{\max} := s_{\max} + \frac{\eta}{\Delta t} (\chi_{\max} - \chi_{\min})$ 
13:   $\lambda := (\lambda_{\min} + \lambda_{\max}) / 2$ 
14:  for  $n_{\text{bisect}} = 1, 2, \dots$  do
15:    for  $E \in \mathcal{T}_h$  do
16:       $\chi_E^{\text{tr}} := \chi_E + \frac{\Delta t}{\eta} (p_E - \lambda + r_E)$ 
17:       $\chi_E^{\text{tr}} := \max \{ \chi_{\min}, \min \{ \chi_{\max}, \chi_E^{\text{tr}} \} \}$ 
18:    end for
19:     $\rho^{\text{tr}} := \frac{1}{|\Omega|} \sum_{E \in \mathcal{T}_h} |E| \chi_E^{\text{tr}}$ 
20:    if  $|\rho^{\text{tr}} - \varrho_0| < \epsilon_{\text{density}}$  then
21:      break
22:    end if
23:    if  $\rho^{\text{tr}} > \varrho_0$  then
24:       $\lambda_{\min} := \lambda$ 
25:    else
26:       $\lambda_{\max} := \lambda$ 
27:    end if
28:     $\lambda := (\lambda_{\min} + \lambda_{\max}) / 2$ 
29:  end for
30:   $\chi := \chi^{\text{tr}}$ 
31: end for

```

For the density update, the regularization term $\nabla \cdot (\beta \nabla \chi)$ needs to be computed. We approximate this term using a cell-centred finite-volume formulation with a

two-point flux approximation (TPFA), following (Droniou, 2014, Section 2). For spatially constant β , the standard Taylor-based TTO treatment approximates the density field locally by a Taylor expansion and reconstructs the Laplacian required to evaluate $\nabla \cdot (\beta \nabla \chi) = \beta \nabla^2 \chi$ Jantos et al. (2019a). Here, we deviate from this treatment for two reasons. First, the element-size dependence in (34) can lead to large differences in β between adjacent elements, so the regularization term can no longer be reduced to $\beta \nabla^2 \chi$. The flux formulation accounts for these differences by treating β as a piecewise constant coefficient in the face transmissibilities. Second, each TPFA face flux requires only the two adjacent cell values, avoiding the wider neighbourhood search and local least-squares reconstruction of the Taylor approach. The reduced neighbourhood simplifies rebuilding the regularization operator on adaptively refined meshes, where collecting the larger Taylor stencils can be time-consuming.

Let $K \in \mathcal{T}_h$ denote a cell with volume $|K|$, barycentre $\mathbf{x}_K$, and density χ_K . Integrating (30) over K and dividing by $|K|$ gives

$$\eta \dot{\chi}_K - p_K + \lambda + \gamma_K - \bar{r}_K \ni 0, \quad \bar{r}_K := \frac{1}{|K|} \int_K \nabla \cdot (\beta \nabla \chi) \, dV. \quad (45)$$

Here, χ_K represents the cell average, $p_K = -|K|^{-1} \int_K p_\chi \, dV$ is the positive driving force defined above, and γ_K is the cell-wise bound reaction. In the discrete problem, the density and the strain obtained from the first-order VEM displacement projection are constant within each element. Consequently, the corresponding elastic driving force is also cell-wise constant. The viscosity η and the volume multiplier λ are spatially constant, while the bound constraint is imposed separately for each cell. Thus, averaging leaves these discrete quantities unchanged; only the regularization term requires information from neighbouring cells. For a higher-order displacement approximation, the nonlinear elastic driving force would instead have to be integrated over the cell.

Introducing the regularization flux $\mathbf{j} := -\beta \nabla \chi$, the divergence theorem yields

$$\bar{r}_K = -\frac{1}{|K|} \sum_{f \in \partial K} Q_{K,f}, \quad Q_{K,f} := \int_f \mathbf{j} \cdot \mathbf{n}_{K,f} \, dA = - \int_f \beta \nabla \chi \cdot \mathbf{n}_{K,f} \, dA. \quad (46)$$

The vector $\mathbf{n}_{K,f}$ is the outward unit normal of cell K on face f , consistent with the boundary-normal notation in (30). The quantity $Q_{K,f}$ is the integrated outward flux, so a positive value contributes to a decrease of χ_K .

For an interior face $f = K|L$, the normal points from K towards its neighbour L .

With $\mathbf{x}_f$ any point in the face plane, define the normal distances

$$\delta_{K,f} := (\mathbf{x}_f - \mathbf{x}_K) \cdot \mathbf{n}_{K,f}, \quad \delta_{L,f} := (\mathbf{x}_L - \mathbf{x}_f) \cdot \mathbf{n}_{K,f}. \quad (47)$$

Both distances are assumed positive, i.e., the cell centres lie on the respective interior sides of the common face. For $\beta_K, \beta_L > 0$, the TPFA flux is

$$Q_{K,f} \approx F_{K,f} := T_f(\chi_K - \chi_L), \quad T_f := \frac{|f|}{\delta_{K,f}/\beta_K + \delta_{L,f}/\beta_L}, \quad F_{L,f} = -F_{K,f}. \quad (48)$$

Here, $|f|$ is the face area, or the edge length in two dimensions. The denominator accounts for the two cell-to-face distances and their respective coefficients. It follows from eliminating a common face value between the two one-sided flux approximations while imposing continuity of the normal flux (Droniou, 2014, Equations (2.2)–(2.4)). At a coarse–fine interface, each active subface is treated separately with its two incident cells. On the exterior boundary, $F_{K,f} = 0$ directly enforces the homogeneous Neumann condition.

Substitution into (45) gives the semi-discrete density evolution

$$\eta \dot{\chi}_K - p_K + \lambda + \gamma_K - \frac{1}{|K|} \sum_{f=K|L} T_f(\chi_L - \chi_K) \ni 0, \quad (49)$$

where the sum runs over the interior faces of K . On an orthogonal mesh, the vector $\mathbf{x}_L - \mathbf{x}_K$ is parallel to $\mathbf{n}_{K,f}$. On a nonorthogonal mesh, the centre-to-centre density difference also contains a contribution from the tangential variation of χ . The TPFA formula omits the corresponding tangential correction and therefore does not, in general, provide a consistent face-flux approximation on such meshes. Its suitability for regularizing the density field on the nonorthogonal meshes considered here remains to be verified numerically.

Collecting the coefficients of the cell densities yields the sparse matrix-vector product

$$\mathbf{r}^{\text{TPFA}} = \mathbf{R}_{\text{TPFA}}(\boldsymbol{\beta})\boldsymbol{\chi}, \quad r_K^{\text{TPFA}} = \frac{1}{|K|} \sum_{f=K|L} T_f(\chi_L - \chi_K). \quad (50)$$

If $\mathcal{N}(K)$ denotes the set of face neighbours of K , the matrix entries are

$$(\mathbf{R}_{\text{TPFA}}(\boldsymbol{\beta}))_{KL} = \begin{cases} \frac{1}{|K|} \sum_{f=K|L} T_f, & L \in \mathcal{N}(K), \\ -\frac{1}{|K|} \sum_{f=K|J} T_f, & L = K, \\ 0, & \text{otherwise.} \end{cases} \quad (51)$$

Thus, each row contains only a diagonal entry and contributions from face neighbours. The row sums vanish, so a constant density field produces no regularization force.

Finally, the common driving-force factor in (34) can be separated from the geometric coefficients. For $p_{\text{avg}} > 0$, write

$$\hat{\beta}_K := \frac{\beta_K}{p_{\text{avg}}} = \max\{\beta^*, h_K^2\}, \quad \mathbf{R}_{\text{TPFA}}(\boldsymbol{\beta}) = p_{\text{avg}} \hat{\mathbf{R}}_{\text{TPFA}}, \quad (52)$$

where $\hat{\mathbf{R}}_{\text{TPFA}}$ is assembled using $\hat{\beta}_K$ in place of β_K . This normalized matrix is constant for a fixed mesh and fixed user parameters and is rebuilt after mesh adaptation. Using $\eta = p_{\text{avg}}\eta^*$, the explicit Euler update with bound projection becomes

$$\chi_K^{i+1} = \Pi_{[\chi_{\min}, \chi_{\max}]} \left[\chi_K^i + \frac{\Delta t}{\eta^*} \left(\frac{p_K^i - \lambda^i}{p_{\text{avg}}^i} + (\hat{\mathbf{R}}_{\text{TPFA}} \boldsymbol{\chi}^i)_K \right) \right]. \quad (53)$$

Here, $\Pi_{[\chi_{\min}, \chi_{\max}]}$ denotes clipping to the admissible density interval and implements the bound reaction γ_K . At each density-update step, the scalar multiplier λ^i is determined by bisection such that $\sum_{K \in \mathcal{T}_h} |K| \chi_K^{i+1} = \varrho_0 |\Omega|$.

3 Adaptive Thermodynamic Topology Optimization

3.1 Refinement Indicators

In this section, we describe the refinement/coarsening technique employed in this work.

As stated in the introduction, marking strategies based purely on the density are widespread. Those approaches can be further separated [into](#) two categories:

1. Approaches which refine elements above a predefined density threshold and coarsen elements with a density below the threshold (cf. Ramm et al. (1998);

Costa Jr and Alves (2003)).

2. Approaches which refine elements in regions with large density gradients and coarsen elements in other regions (cf. Stainko (2005); Lambe and Czekanski (2018); Vogel and Junker (2021)).

Techniques belonging to the first category come with the benefit of a very finely resolved structure and therefore also a fine mesh at regions of high stresses. The obvious drawback is the still rather large number of elements required. Especially for problems where a large fraction of the design area is filled with material.

Techniques which do belong to the second category ideally require fewer elements, since only the boundary of the structure is highly resolved. This is offset by the fact that an accurate computation of the displacement variable is not assured. The importance to account for the displacement error will be shown in Section 4.3. Only a minority of papers takes the displacement error into account for the refinement/coarsening marking (Costa Jr and Alves (2003); Wang et al. (2014); Salazar De Troya and Tortorelli (2018)).

Existing techniques can be further categorized by the frequency the refinement takes place and whether a fine or coarse mesh is chosen at the beginning of the optimization.

The approach with the lowest computational cost only applies AMR when the topology optimization has converged. Doing so, the main structural features are determined by the initial mesh and only the sharpness of the void/bulk interfaces is increased (cf. Ramm et al. (1998); Costa Jr and Alves (2003); Stainko (2005); Wang et al. (2014); Lambe and Czekanski (2018)).

An alternative is to start with a coarse mesh and adapt the mesh after each optimization step, with the goal to search the full design space to find the ideal structural design. Depending on the problem, this is not always sufficient to obtain the same results as if a constantly fine mesh would have been used (cf. Wang et al. (2010); Zhang et al. (2020a); Vogel and Junker (2021); Herrero-Pérez et al. (2022); Cihan et al. (2024)).

A remedy is to start the optimization already with the finest mesh and to refine/coarsen afterward (only done in Herrero-Pérez et al. (2022)).

The marking strategy proposed in this work is a combination of a design based marking and an error based marking. At the start of the optimization, we refine the coarse mesh up to the finest level and apply AMR after each optimization step.

3.1.1 Marking Strategy

In this section, we describe the marking strategy employed in this work. For the error-based marking, different versions of the presented DWR method are utilized, which differ in how the analytical adjoint function is approximated. As goal functional we employ $J(\mathbf{u}) = \frac{1}{2}A(\mathbf{u}, \mathbf{u}) = \frac{1}{2} \int_{\Omega} \chi^3 \boldsymbol{\varepsilon} : \mathbb{C}_0 : \boldsymbol{\varepsilon} \, d\Omega$. Here, A and S include the elementwise density factor χ^3 , which is held fixed during error estimation. Using this goal functional, the problem is self-adjoint, which means $\mathbf{u} = \mathbf{z}$. We use $\mathbf{u}_h = \mathbf{z}_h$. By projection orthogonality, $J(\mathbf{u}_h) - J_h(\mathbf{u}_h) = \frac{1}{2}S^{\text{corr}}(\mathbf{u}_h, \mathbf{z}_h)$. Since the problem is self-adjoint, (18a) gives

$$\begin{aligned} J(\mathbf{u}) - J_h(\mathbf{u}_h) &= \frac{1}{2} \left(F(\mathbf{e}^*) - A(\mathbf{u}_h, \mathbf{e}^*) \right) \\ &\quad + \frac{1}{2} \underbrace{\left(D_{\mathbf{u}}J(\mathbf{u}_h)[\mathbf{e}] - A(\mathbf{e}, \mathbf{z}_h) \right)}_{=0} \\ &\quad - \frac{1}{2}S^{\text{corr}}(\mathbf{u}_h, \mathbf{z}_h). \end{aligned} \tag{54}$$

Here, it can be seen that the second term of the error identity equals zero.

For this first DWR version we approximate¹ $\mathbf{z}$ as $\mathbf{z}_h^{(2)}$, which denotes the solution of the adjoint problem with quadratic shape functions. This technique comes with a high additional cost and will therefore only serve as reference solution.

For the second approach we use the GPRMSC method (cf. Sellmann and Junker (2025)) to approximate the adjoint function $\mathbf{z}$. The basic idea of the GPRMSC is to reconstruct a higher-order solution by treating the values at additional mid-edge/mid-element nodes (required for quadratic approximations) as unknowns while preserving the original nodal values from the linear solution. Additionally, the internal moments are condensed, applying static condensation. This does result in a much smaller system of equations

$$\mathbf{K}^{\text{rec}} \cdot \hat{\mathbf{z}}_h^{\text{rec}} = \hat{\mathbf{f}}^{\text{rec}} \tag{55}$$

where $\mathbf{K}^{\text{rec}}$ denotes the reconstruction stiffness matrix, $\hat{\mathbf{z}}_h^{\text{rec}}$ the reduced vector of unknowns and $\hat{\mathbf{f}}^{\text{rec}}$ the resulting forces on the remaining degrees of freedom. In Sellmann and Junker (2025), it was found for the Poisson equation that the condition number of the reconstruction matrix is extremely small (around 5 - 30). Consequently, the linear system of equations can be solved with just a few iterations of an iterative solver. For our implementation, we employed the conjugate gradient

¹For clarity, we still denote the adjoint function as $\mathbf{z}$, despite the equality $\mathbf{z} = \mathbf{u}$

solver implemented in Montoison and Orban (2023) to solve the resulting system of equations. Due to the small possible values of χ^3 the condition of the problem is strongly deteriorated. As a remedy, a simple scaling (similar to mass lumping) with the diagonal matrix

$$\mathbf{T}[i, j] = \begin{cases} \frac{1}{\sqrt{\mathbf{K}^{\text{rec}}[i, j]}} & \text{if } i = j \\ 0 & \text{otherwise} \end{cases} \quad (56)$$

is applied. Here, $\mathbf{T}[i, j]$ denotes the entry of $\mathbf{T}$ in the i -th row and j -th column. We then solve the preconditioned system $\mathbf{T} \cdot \mathbf{K}^{\text{rec}} \cdot \mathbf{T} \cdot \mathbf{y} = \mathbf{T} \cdot \hat{\mathbf{f}}^{\text{rec}}$ for $\mathbf{y}$ and recover $\hat{\mathbf{z}}_h^{\text{rec}} = \mathbf{T} \cdot \mathbf{y}$. Since $\mathbf{T}$ is diagonal, this preconditioning adds negligible computational cost and does yield condition numbers around 20 in our numerical experiments.

For the last approach we use $\mathbf{z} \approx \mathbf{z}_h^\pi$. In two dimensions, a nonzero indicator is possible only for elements with more than 3 nodes, since $\mathbf{z}_h = \mathbf{z}_h^\pi$ on triangles. We split the load functional into volume and traction contributions, $F = F_\Omega + F_\Gamma$, where

$$F_\Omega(\mathbf{v}) := (\mathbf{f}, \mathbf{v})_\Omega, \quad F_\Gamma(\mathbf{v}) := (\mathbf{t}, \mathbf{v})_{\partial\Omega_N}.$$

Proposition 1. *Let $\mathbf{f} \in [H^1(\Omega)]^{N_D}$ and $\mathbf{z} \in [H^2(\Omega)]^{N_D}$ on a shape-regular mesh. If we apply the approximation $\mathbf{z} \approx \mathbf{z}_h^\pi$, the error identity*

$$J(\mathbf{u}) - J_h(\mathbf{u}_h) = \frac{1}{2} \left(F(\mathbf{e}^\star) - A(\mathbf{u}_h, \mathbf{e}^\star) \right) - \frac{1}{2} S^{\text{corr}}(\mathbf{u}_h, \mathbf{z}_h) \quad (57)$$

simplifies, after neglecting the higher-order volume-load contribution, to

$$J(\mathbf{u}) - J_h(\mathbf{u}_h) \approx \frac{1}{2} S(\mathbf{u}_h, \mathbf{z}_h) - \frac{1}{2} F_\Gamma(\mathbf{z}_h - \mathbf{z}_h^\pi). \quad (58)$$

Proof. We first examine the volume contribution:

$$F_\Omega(\mathbf{z}_h - \mathbf{z}_h^\pi) = (\mathbf{f}, \mathbf{z}_h - \mathbf{z}_h^\pi)_\Omega = (\mathbf{f} - \mathbf{f}^\pi, \mathbf{z}_h - \mathbf{z}_h^\pi)_\Omega. \quad (59)$$

Here, $\mathbf{f}^\pi$ denotes the elementwise L^2 projection onto $[\mathbb{P}_1]^{N_D}$, and we used the projection property (8). Next, we apply the Cauchy-Schwarz inequality which yields:

$$|(\mathbf{f} - \mathbf{f}^\pi, \mathbf{z}_h - \mathbf{z}_h^\pi)_\Omega| \leq \|\mathbf{f} - \mathbf{f}^\pi\|_{L^2} \|\mathbf{z}_h - \mathbf{z}_h^\pi\|_{L^2}. \quad (60)$$

Using the local Poincaré inequality for zero mean functions (cf. Brenner and Scott (2008)), the projection estimates and the approximation property of the VEM discretization $|\mathbf{z} - \mathbf{z}_h|_{H^1} \leq Ch|\mathbf{z}|_{H^2}$ (cf. Beirão Da Veiga et al. (2023)), we can

bound the two L^2 -norms:

$$\begin{aligned} \|\mathbf{f} - \mathbf{f}^\pi\|_{L^2} \|\mathbf{z}_h - \mathbf{z}_h^\pi\|_{L^2} &\leq C_1 h^2 |\mathbf{f}|_{H^1} |\mathbf{z}_h - \mathbf{z}_h^\pi|_{H^1} \\ &\leq C_2 h^3 |\mathbf{f}|_{H^1} |\mathbf{z}|_{H^2}. \end{aligned} \quad (61)$$

Sobolev seminorms of piecewise functions are understood elementwise. The last inequality uses the stability and approximation property of the projection together with the stated VEM error bound. If $\mathbf{f} \in [H^2(\Omega)]^{N_D}$, the volume contribution converges with $\mathcal{O}(h^4)$. Since this error contribution is of order $\mathcal{O}(h^3)$, which is one order higher than the expected quadratic order of convergence of the strain energy error, we neglect it in this approximation. The traction contribution F_Γ is retained, since the volume estimate does not apply to it.

Using the projection orthogonality (7a), we have $A(\mathbf{u}_h, \mathbf{z}_h - \mathbf{z}_h^\pi) = A(\mathbf{u}_h - \mathbf{u}_h^\pi, \mathbf{z}_h - \mathbf{z}_h^\pi)$. Consequently, substituting $\mathbf{e}^* \approx \mathbf{z}_h^\pi - \mathbf{z}_h$ into (54) and using the definition of $S^{\text{corr}}(\mathbf{u}_h, \mathbf{z}_h)$, we obtain:

$$\begin{aligned} J(\mathbf{u}) - J_h(\mathbf{u}_h) &\approx \frac{1}{2} \left(F(\mathbf{z}_h^\pi - \mathbf{z}_h) - A(\mathbf{u}_h, \mathbf{z}_h^\pi - \mathbf{z}_h) \right) \\ &\quad - \frac{1}{2} S^{\text{corr}}(\mathbf{u}_h, \mathbf{z}_h) \\ &= \frac{1}{2} S(\mathbf{u}_h, \mathbf{z}_h) - \frac{1}{2} F(\mathbf{z}_h - \mathbf{z}_h^\pi) \\ &\approx \frac{1}{2} S(\mathbf{u}_h, \mathbf{z}_h) - \frac{1}{2} F_\Gamma(\mathbf{z}_h - \mathbf{z}_h^\pi). \end{aligned} \quad (62)$$

□

In the present self-adjoint case, the stabilization part is related to the displacement-based indicator proposed in van Huyssteen, Daniel et al. (2022), which measures the nodal deviation from a linear displacement field. The estimator additionally retains the traction contribution $F_\Gamma(\mathbf{z}_h - \mathbf{z}_h^\pi)$. Only the polynomial projection, stabilization and boundary values are required to evaluate the estimator. No additional adjoint problem needs to be solved. In the following we will refer to the first approach as DWR based marking, the second one as recDWR (reconstruction DWR) based marking and the last one as sDWR (simplified DWR) based marking. After computing the element-wise error indicator the following marking strategy is applied. Let n_c denote the number of children an element is split into when refined. Further, η_E denotes the indicator of element E . We first compute the mean error $\bar{\eta}$ across all elements. We then mark elements for refinement when $\eta_E > 2\bar{\eta}$ and for coarsening when $\eta_E < \frac{2}{n_c^{1+2/N_D}} \bar{\eta}$. These bounds are based on the quadratic convergence of the error estimator w.r.t. the element diameter. When an element

Algorithm 2 Density Evolution AMR Strategy

```

1: for each optimization step do
2:   Compute displacement field  $\mathbf{u}_h$  (13)
3:   Update the density using Algorithm 1 and record  $\Delta\chi_E$  for each element
4:   for  $E \in \mathcal{T}_h$  do
5:     Compute  $\eta_E$  using one of the described DWR versions ((54) or (58))
6:   end for
7:   Compute mean error  $\bar{\eta}$  across all elements
8:   for  $E \in \mathcal{T}_h$  do
9:     if  $\eta_E > c_1\bar{\eta}$  with  $c_1 = 2.0$  or  $\Delta\chi_E > 0$  then
10:      if Element refinement level < max refinement level then
11:        Mark element for refinement
12:      end if
13:    else if  $\eta_E < c_2\bar{\eta}$  with  $c_2 = 2/n_c^{1+2/N_D}$  and  $\Delta\chi_E = 0$  then
14:      if Element refinement level > 1 then
15:        Mark element for coarsening
16:      end if
17:    end if
18:  end for
19:  Apply refinement/coarsening
20:  Proceed with optimization step
21: end for

```

with an error $2\bar{\eta}$ is split, each child has roughly an error of $\eta_E/n_c^{1+2/N_D}$ (the error sum of the children is $\eta_E/n_c^{2/N_D}$ which aligns with the expected convergence rate) (c.f. van Huyssteen, Daniel et al. (2022)). This brings the element-wise error back into the interval of acceptable errors. If the factor between the upper and lower refinement bound is chosen smaller, oscillations of the number of nodes and the strain energy occur in the convergence process.

In addition, elements are marked based on the net density change $\Delta\chi_E := \chi_E^{\text{new}} - \chi_E^{\text{old}}$ over the complete density update within an optimization step. An element E is marked for refinement if $\Delta\chi_E > 0$, while coarsening requires $\Delta\chi_E = 0$ in addition to the error criterion. This definition comes with two advantages. First, no additional hyperparameters are required. Second, the mesh quickly adapts to capture changes in the density field. The full algorithm can be found in Algorithm 2.

3.2 Regularization and Pseudo Laplacian

As described in [Vogel and Junker \(2021\)](#), the regularization parameter β (34) needs to be chosen locally to avoid checker-boarding. If not stated differently we use $\beta_E = p_{\text{avg}} \max(h_E^2, \beta^*)$. Because the maximum refinement level of adjacent elements can differ greatly, oscillations of the compliance are observed during the optimization. This issue arises from how elements at structure-void interfaces are handled and how their regularization is applied.

For instance, consider a small element placed next to a larger element. The density variable evolves based on [both elements' states](#) and the distance between their barycenters. When the larger element is refined, each child element inherits the parent's state. As a result, during the computation of the Laplacian, the neighboring elements have the same states as before but lie closer together. This then leads to significant changes of the Laplacian, causing the observed oscillations. Similar effects do occur when elements are coarsened.

To address these issues, we propose a modified pseudo-Laplacian, based on re-scaled distances between element barycenters. The idea is to scale the adjacent element size so that it matches the size of the current element. For the computation of the Laplacian we then use the distance between the barycenter of the current element and its scaled neighbors. By adjusting the effective distance proportionally to the size ratio, the influence of differently sized elements is normalized. The re-scaled distance vector between two adjacent elements is computed as:

$$\mathbf{d}_{1,0} = [\Delta\bar{x}, \Delta\bar{y}]^T = \left(\mathbf{x}_1^{\text{bc}} - \mathbf{x}_0^{\text{bc}} \right) \frac{h_0}{\text{mean}(h_1, h_0)} \quad (63)$$

where $\mathbf{x}_i^{\text{bc}}$ represents the barycenter coordinates of element i , and h_i denotes the respective element diameters. Then, $\Delta\bar{x}$ and $\Delta\bar{y}$ replace Δx and Δy in the equations (35) and (36) based on which the Laplacian is computed.

4 Numerical Results

In this section our proposed refinement strategy is tested using different benchmark cases. First, we show the overall performance of the proposed method, comparing the final structure as well as the convergence behavior with the uniformly refined mesh. Here, the sDWR method is utilized for the error computation.

Second, we compare the sDWR method with the more accurate but also more expensive DWR and recDWR methods. A qualitative comparison of the computational cost is not performed since our recDWR and DWR frameworks are not yet optimized.

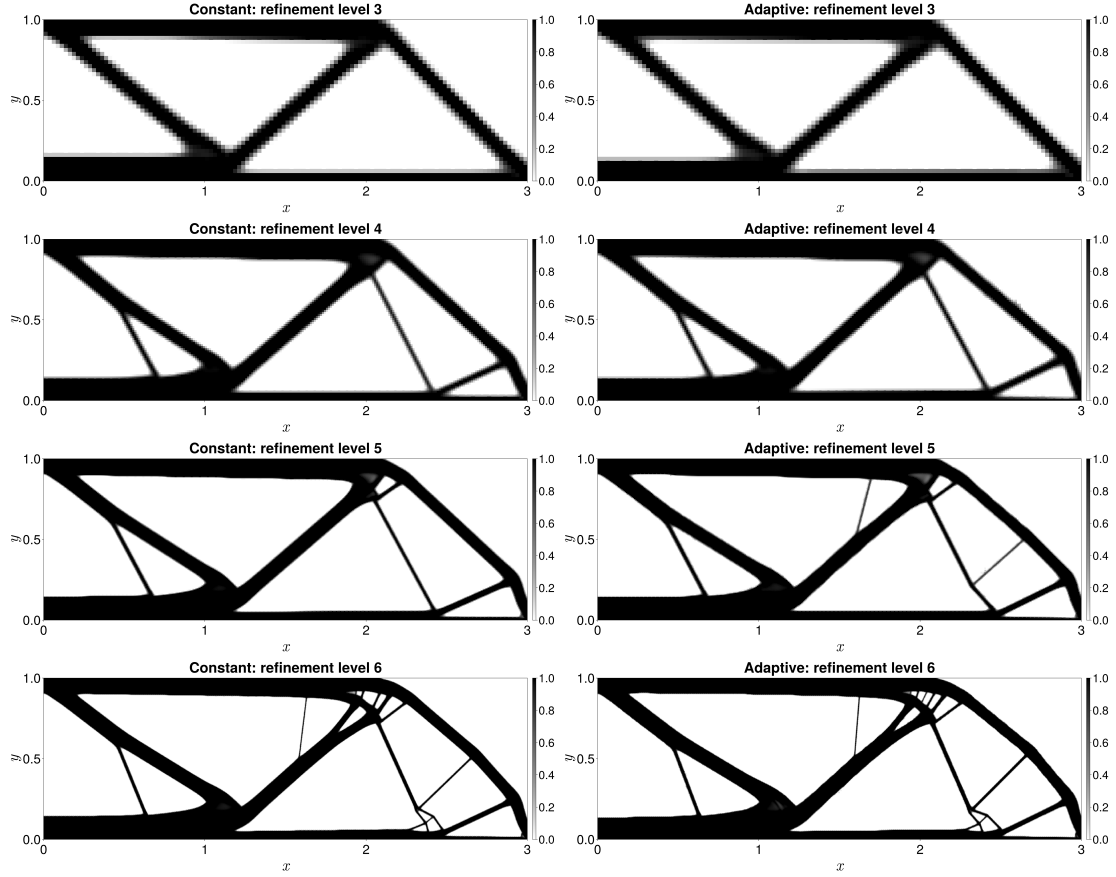


Figure 3: Comparison of the Density field χ for the MBB Beam for different refinement levels after 500 Iterations

Third, we show the influence of the incorporation of the displacement error into the refinement marking strategy and finally we investigate the necessity to utilize the pseudo Laplacian. An overview of the simulation parameters can be found in table 1.

4.1 MBB-beam: general performance assessment

For the first benchmark case, the well known MBB-beam, which is depicted in Figure 4, is considered.

Symbol	Quantity	Value
$\chi_{\min}$	Minimal density	$10^{-3} [-]$
E	Young's modulus	210 [MPa]
ν	Poisson ratio	0.33 $[-]$
ϱ_0	Target relative density	0.3 $[-]$
β^*	Regularization parameter	$h_{\min}^2 [-]$
η^*	Viscosity	15.0 [1/s]
Δt^*	Time step	1.0 [s]

Table 1: Simulation parameters

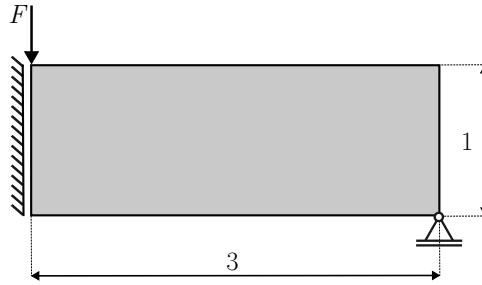


Figure 4: Geometry of the MBB-beam

We compare the final optimized structure after 500 iterations for the adaptive and constant mesh for different maximum refinement levels (see Figure 3). The coarsest mesh is referred to as refinement level 1 which consists of 30 elements in x -direction and 10 elements in y -direction. Consequently, the adaptive mesh can consist of elements which have a size between the coarsest mesh and the finest allowed element size.

Comparing the final structures (see Figure 3), the outcomes are qualitatively the same. Only for the coarsest mesh, the outcomes differ. The structure of the adaptive mesh contains one more diagonal structure part. However, this part also dissolves if the optimization is run for 600 iterations in total.

To obtain a more detailed comparison, we plot the density difference between the adaptive mesh and the constant mesh exemplary for a maximum refinement level of 6. To do so, we refine all elements of the adaptive mesh after the optimization up to the finest refinement level. All new elements get the state of their respective parent.

The comparison is depicted in Figure 5. It can be seen that only the states at the structure-void interfaces differ, which underlines the good agreement between the final structures. For different maximum refinement levels, similar results are

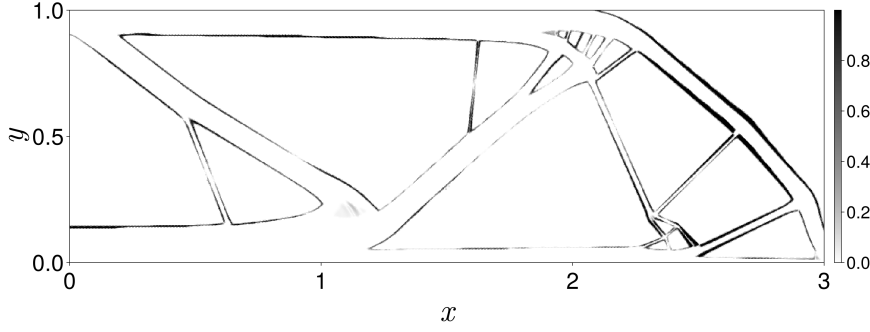


Figure 5: Absolute state difference $|\chi_{\text{const}} - \chi_{\text{adapt}}|$ for a maximum refinement level of 6 after 500 iterations.

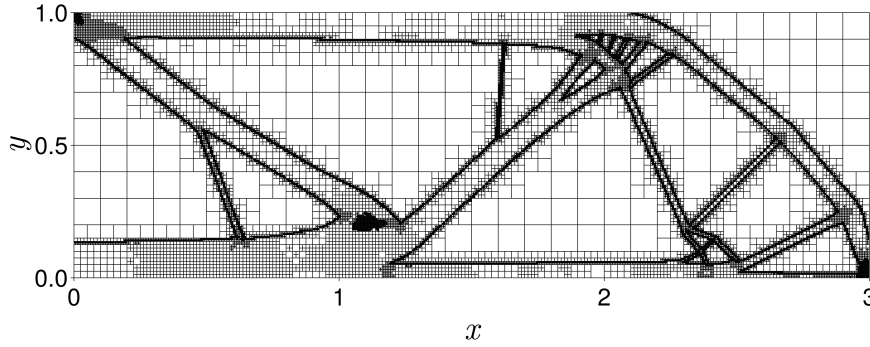


Figure 6: Adaptively refined mesh after 500 iterations.

obtained. The occurring differences are unavoidable. Due to the different element sizes the values of the Laplacian are different at these locations and in consequence also the transition layer differs.

An exemplary picture of the adaptively refined mesh for a maximum refinement level of 6 after 500 iterations can be found in Figure 6. The picture of the evolution of the grid alongside the state is depicted in A Figure 16.

When investigating the properties of the density fields during the optimization process, only small differences occur (see Figure 7a). Here, we utilized the metric measure of non-discreteness (MOD) (cf. Sigmund (2007)) which is defined as

$$\text{MOD} := \frac{4}{|\Omega|} \int_{\Omega} (\chi - \chi_{\min})(\chi_{\max} - \chi) \, d\Omega. \quad (64)$$

The smaller the MOD, the sharper the void-structure interfaces. The largest result difference occurs for a maximum refinement level of 3 due to the reason explained

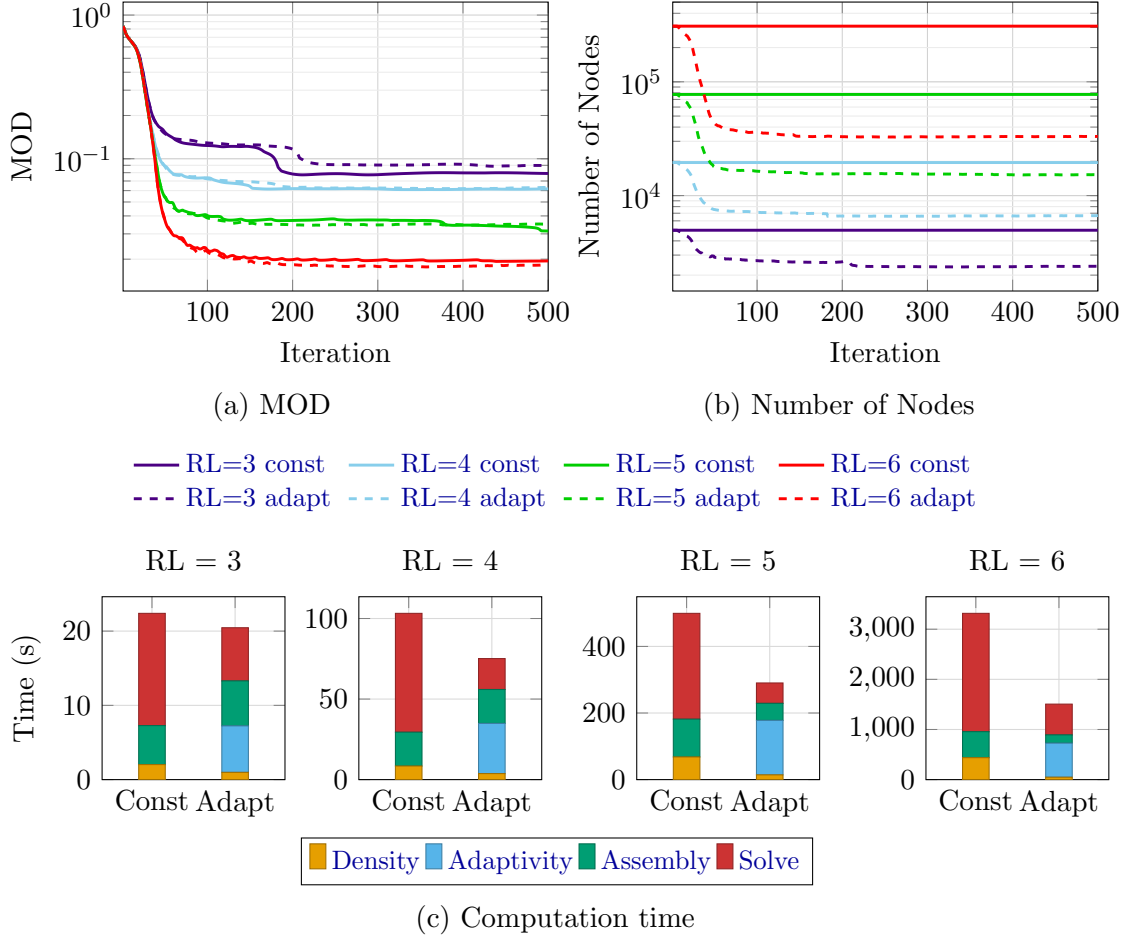


Figure 7: Comparison of the optimization process for the MBB-beam.

beforehand. The main advantages of the adaptive scheme employed do become prominent when comparing the computation time required for the optimization showing a difference of a factor of approximately 2.2 for the finest mesh. The comparison of two further benchmark cases can be found in Appendix A where also a strong agreement between the final structures can be observed. This highlights the validity and robustness of the proposed scheme.

4.2 MBB-beam: Comparison of the DWR schemes

In the following, we compare the full DWR scheme with its simplified counterparts. [The main purpose of this section is to justify the use of the sDWR method.](#) Nevertheless, the recDWR method is discussed alongside as a tradeoff between

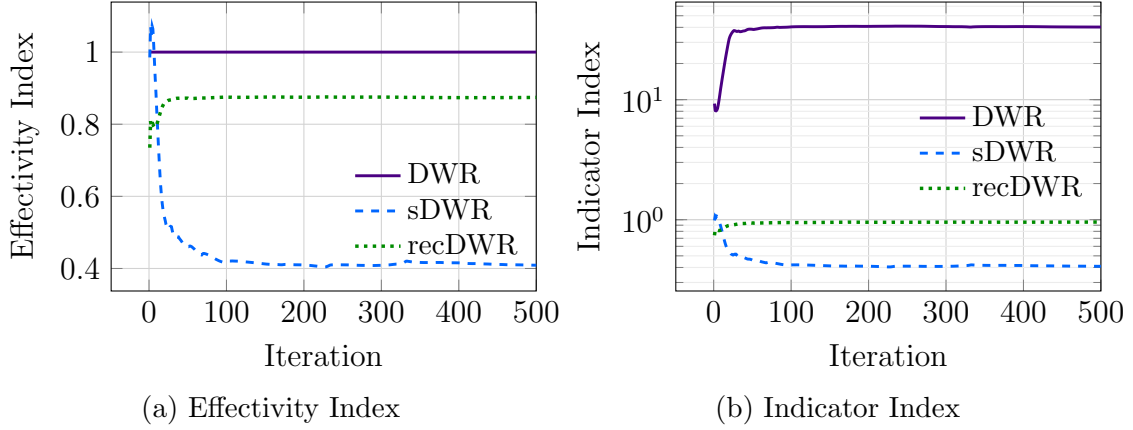


Figure 8: Effectivity and Indicator Index on an adaptively refined mesh with a maximum refinement level of 5 for different approximations of the analytical adjoint function.

accuracy and computational cost. But due to the high time required to compute the local stiffness matrices for quadratic shape functions in our current implementation, this approach is still infeasible for its application in the TTO framework. As a benchmark case, we again consider the MBB-beam. For the remainder of this section, in order to evaluate the error identity (54), we replace the singular external force with a traction force distributed over the interval $x \in (0, 0.15)$. First, we compare the indicator and effectivity index for all three methods. The results are presented for a refinement driven by a full DWR based error indicator. As reference solution we identify $\tilde{J} = J_h^{(2)}(\mathbf{u}_h^{(2)})$, where $J_h^{(2)}$ uses the quadratic VEM bilinear form.

In Figure 8, both the effectivity and indicator index are depicted. The effectivity index of the full DWR method remains almost exactly one over all iterations, which is expected for a quadratic reference solution but verifies the correctness of the scheme. At the start of the optimization, the sDWR does outperform the recDWR method, reaching an effectivity and indicator index close to one. However, as the optimization proceeds the performance of the sDWR method deteriorates, reaching an effectivity index of around 0.6 compared to the value of 0.8 obtained using the recDWR method.

Another important observation is the high indicator index above 15, when the DWR method is used. Both other methods result in values between 0.5 and 0.9. So the absolute sum of all element error contributions exceeds the overall error by far if the most accurate approximation of the adjoint function is used which strongly decreases the suitability of the DWR scheme for adaptive mesh refinement.

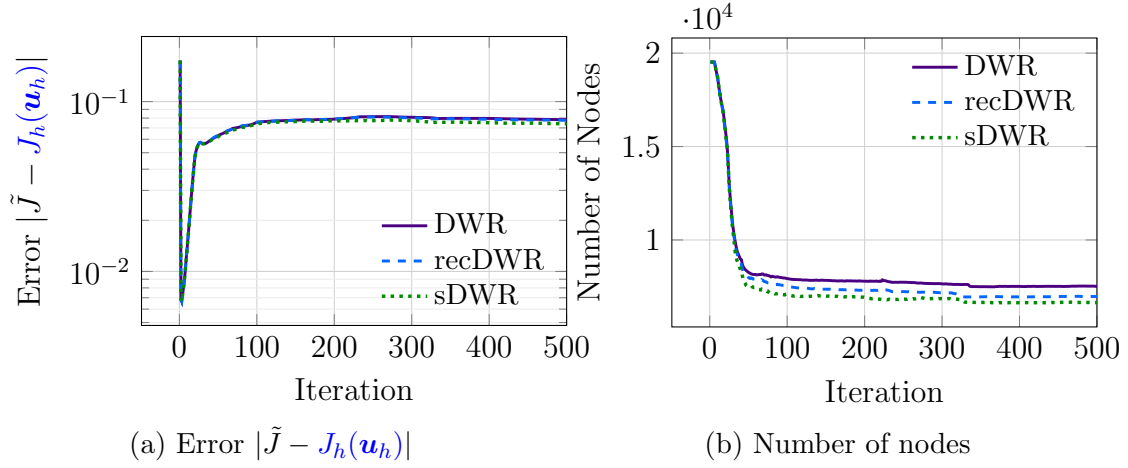


Figure 9: Comparison of the error computing the strain energy and the number of nodes for the MBB-beam for a maximum refinement level of 4.

Qualitatively similar observations are made for finer and coarser meshes and different benchmark cases.

To measure the capability of the different DWR versions to drive the AMR procedure, we compare the discretization error for the adaptively refined meshes and the number of nodes over the iterations. In Figure 9 it can be seen that the number of nodes of the adaptively refined mesh is much smaller for the sDWR method compared with the full DWR method. The number of nodes for the recDWR method is somewhere in between. The resulting fine mesh when using the DWR scheme is most likely attributed to the high indicator index. Interestingly the discretization error is the smallest if the refinement is driven by the sDWR method. However, the small difference might also originate from a decreased accuracy of the quadratic reference solution as the number of nodes is the smallest for the sDWR method.

4.3 L-Cantilever: Influence of the error estimator

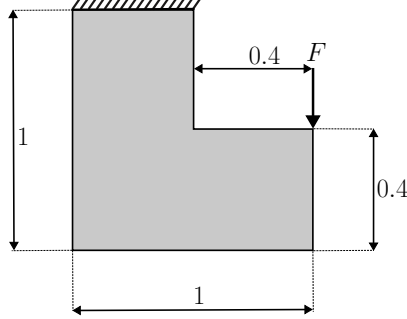


Figure 10: Geometry of the L-Cantilever

To underline the necessity of considering the element error in the refinement process, we compare our results with a purely density-based refinement rule. For this purpose, we use exemplarily the refinement scheme used in [Vogel and Junker \(2021\)](#); [Cihan et al. \(2024\)](#) without limiting the maximal refinement level of adjacent elements². For the marking strategy, we introduce the values $\chi_{\min}^{\text{diff}}$ and $\chi_{\max}^{\text{diff}}$, which denote the smallest and largest absolute state differences, respectively, between the i -th element and its neighbors. Neighbors are elements that share at least one node with the i -th element. Elements are marked for refinement if

- $\chi_{\max}^{\text{diff}} \geq 0.4$ **or** $\chi_E \in (0.25, 0.75)$

and for coarsening if

- $\chi_{\max}^{\text{diff}} \leq 0.05$ **and** $\chi_E \notin (0.1, 0.9)$.

To evaluate the error, we again compare the strain energy between the linear and quadratic solution. For this purpose we compute the relative element-wise error

$$\Psi_E^{\text{rel}} = \frac{|J_{h,E}^{(2)}(\mathbf{u}_h^{(2)}) - J_{h,E}(\mathbf{u}_h)|}{|J_{h,E}^{(2)}(\mathbf{u}_h^{(2)})|}. \quad (65)$$

Here, $J_{h,E}(\mathbf{v}_h) := \frac{1}{2}A_{h,E}(\mathbf{v}_h, \mathbf{v}_h)$ denotes the element contribution to J_h , and the superscript (2) refers to the quadratic VEM approximation. As benchmark example, we utilize the L-shaped cantilever as depicted in Figure 10 and a maximum refinement level of 6. We refer to the marking strategy based on both the change

²We do not simply use the scheme proposed in this work without the error estimator, since this approach leads to large corners in the structure which in return yields large concentration of stresses at these locations.

in the density field and the element-wise error (using the sDWR method) as the mixed marking strategy, while the marking strategy purely based on the density field is termed the density marking strategy. The element-wise strain energy error is depicted in Figure 11a and 11c. The maximal relative error using the density marking exceeds 100% in multiple locations while remaining below 28% for the mixed marking strategy. These differences can also be seen when comparing the sum of element-wise error contributions. Here, the density marking error is four times larger than the global error from the mixed marking strategy. The maximum for the mixed marking strategy is located at the external singular force where the mesh cannot be refined further.

These observations strongly highlight the need for a marking strategy which also considers the computational error.

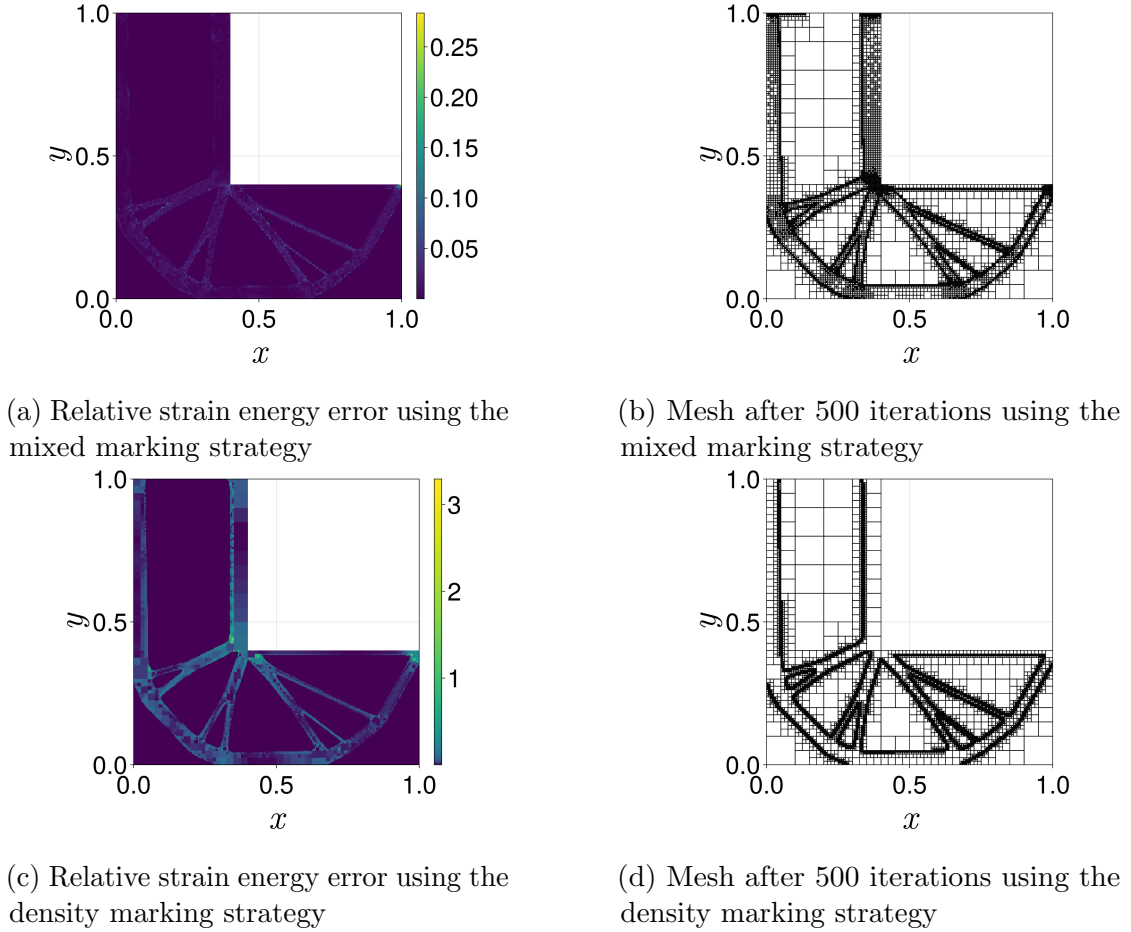


Figure 11: Comparison of the relative strain energy error Ψ_E^{rel} (65) and the final mesh for the density and mixed marking strategy for the L-shaped cantilever

5 Conclusion and Outlook

In this work, we have successfully introduced a novel and robust marking strategy for adaptive topology optimization. The proposed strategy considers both the design variable and the displacement error via DWR, and we demonstrated that incorporating the displacement error is crucial for achieving a reliable numerical scheme. For specific use cases where high accuracy of the numerical scheme is crucial, it might be beneficial to utilize a full DWR scheme. However, one must consider whether the additional accuracy justifies the remarkable extra computational cost. We have also shown the necessity of adapting the standard regularization scheme used in the TTO to obtain a stable convergence behavior, even in the presence of large size differences of adjacent elements. We verified the robustness of our approach through three commonly used benchmark cases. In each of these, the final structure obtained using the adaptive scheme is nearly identical to that obtained with a uniform fine mesh (except for slight differences at the structure–void interfaces). Consequently, adopting an adaptive mesh for topology optimization is highly advantageous due to the significant computational speed up.

In future studies, the proposed scheme could be improved further by:

- Investigating whether the mesh needs to be adapted after every optimization step, particularly when the optimization is nearly converged.
- Determining suitable upper and lower error bounds for refinement to achieve an optimal trade-off between reliability and the required number of elements.

Finally, one could test the proposed scheme for nonlinear material behavior, where the adaptivity could be even more beneficial due to the higher cost of the numerical solver.

6 Numerical Results 3D

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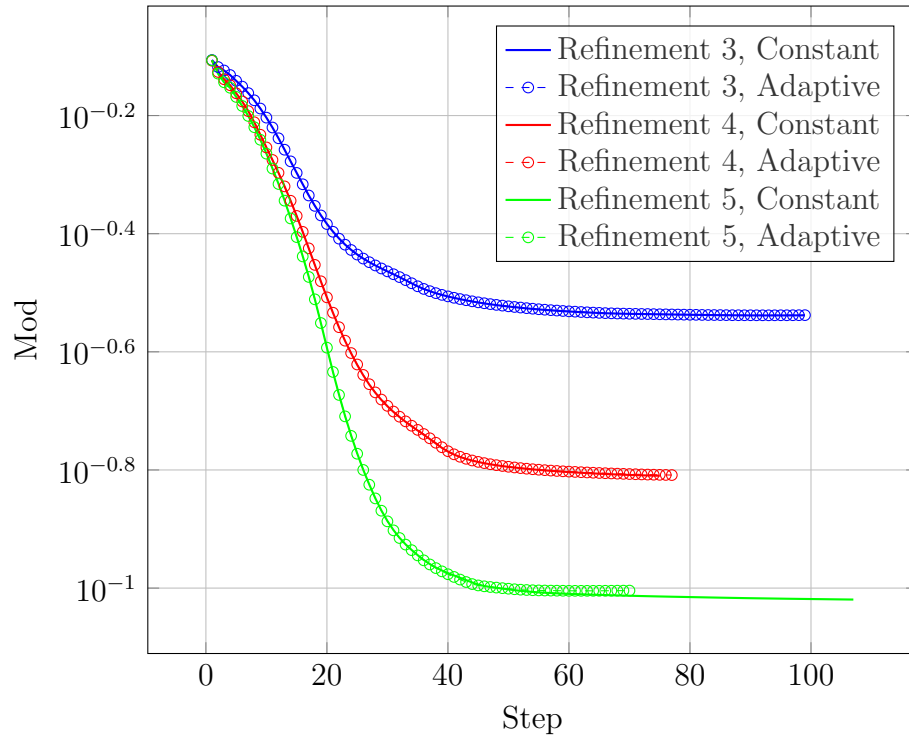


Figure 12: Evolution of Mod over steps

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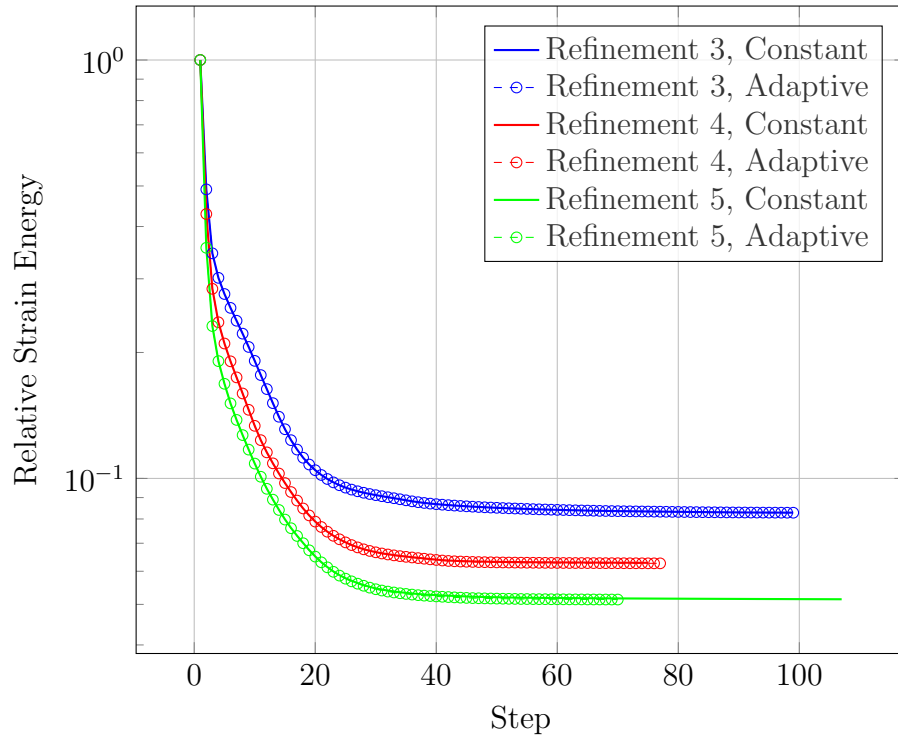


Figure 13: Evolution of Relative Strain Energy over steps

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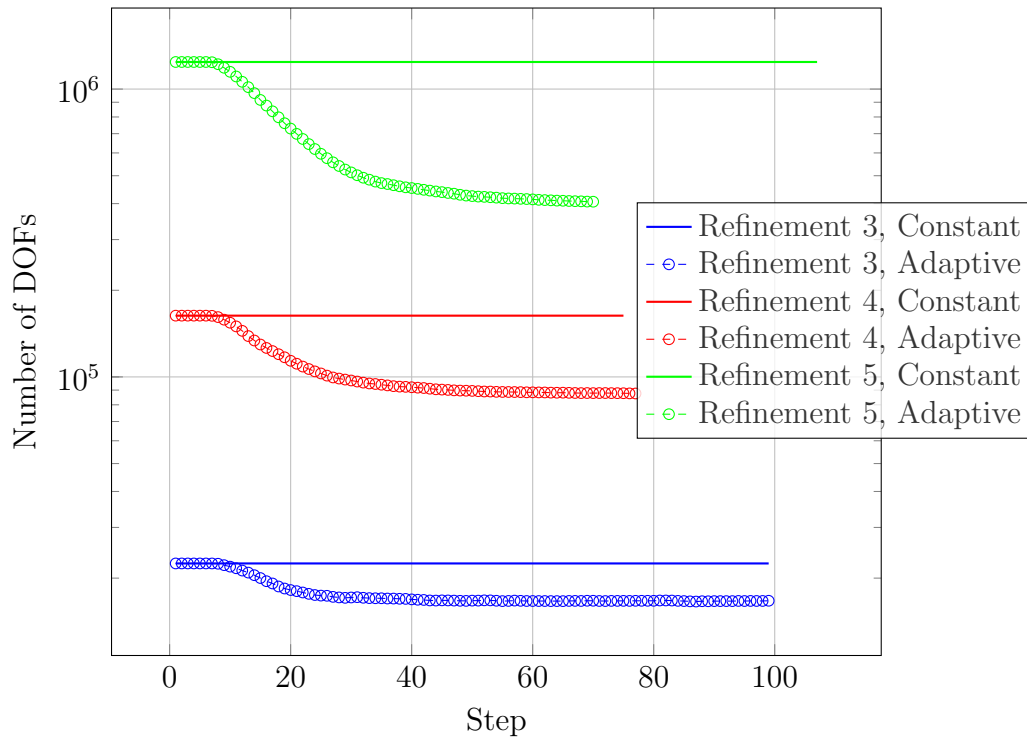


Figure 14: Evolution of Number of DOFs over steps

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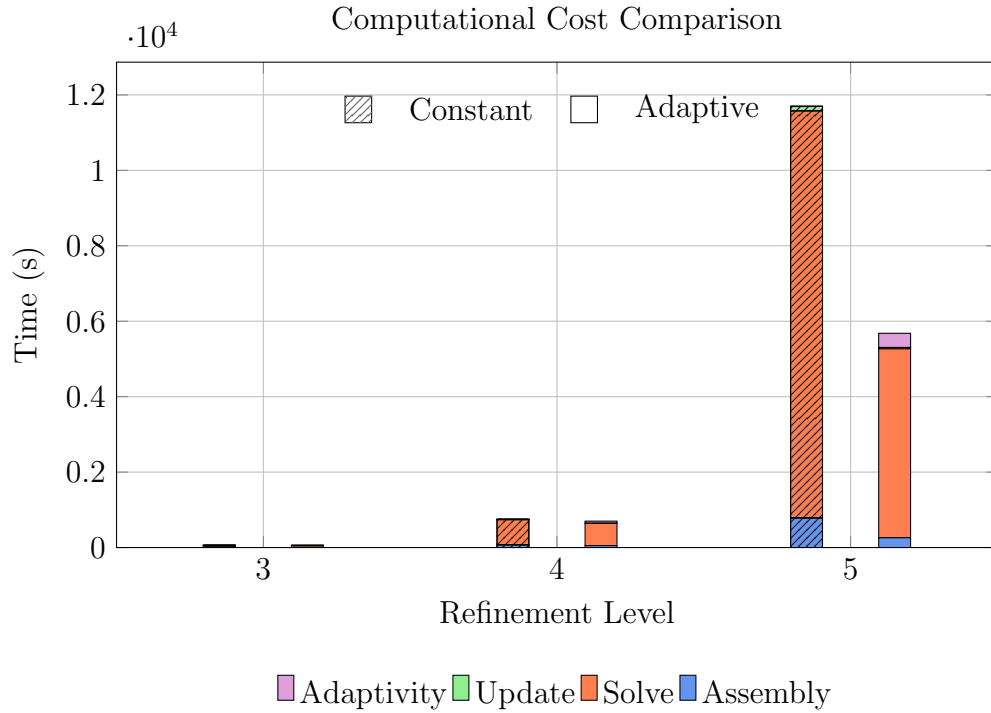


Figure 15: Computational times per refinement level.

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A Appendix A

A.1 Grid and State evolution

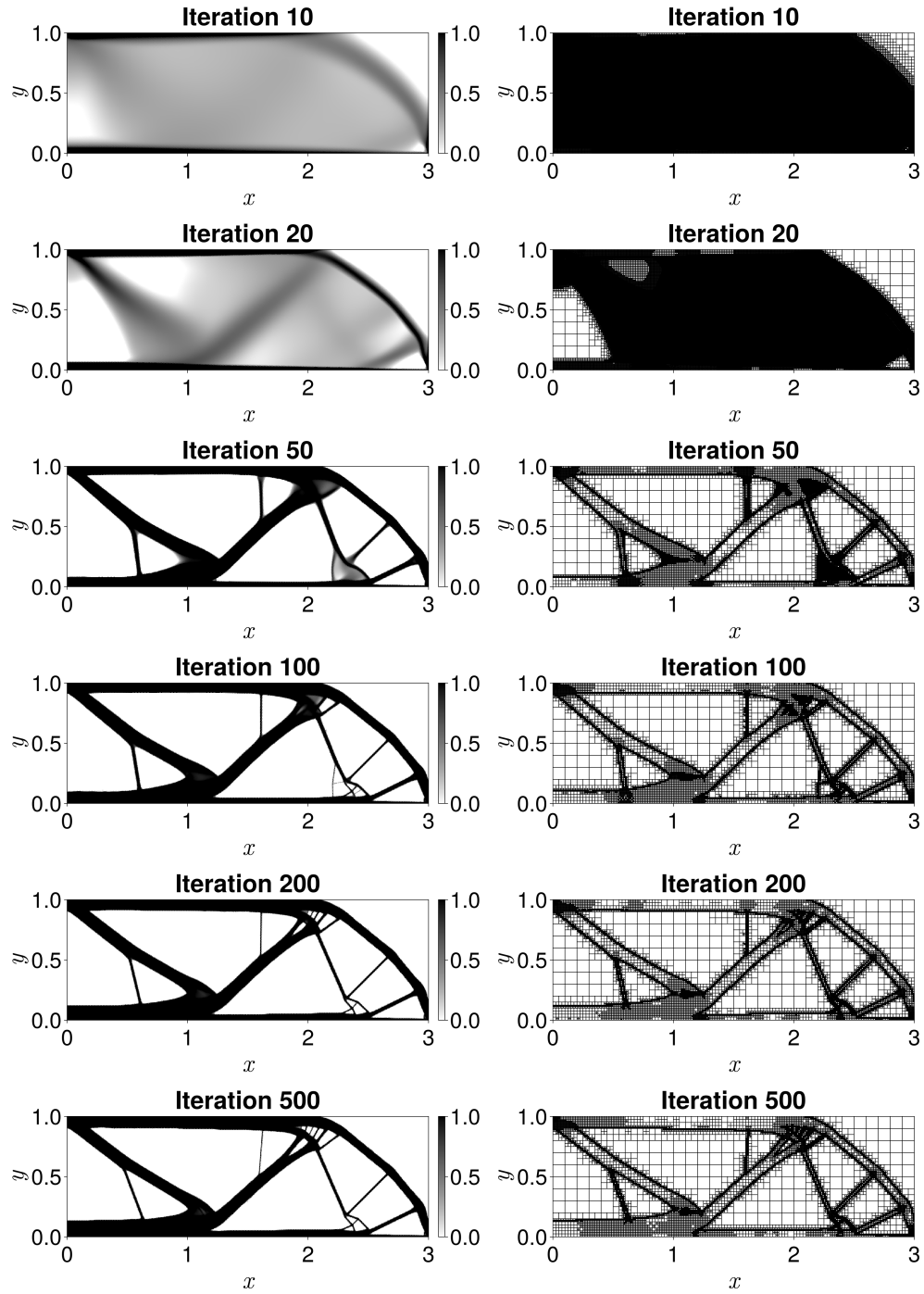


Figure 16: Evolution of the states alongside the mesh for different iterations

A.2 L-shaped cantilever

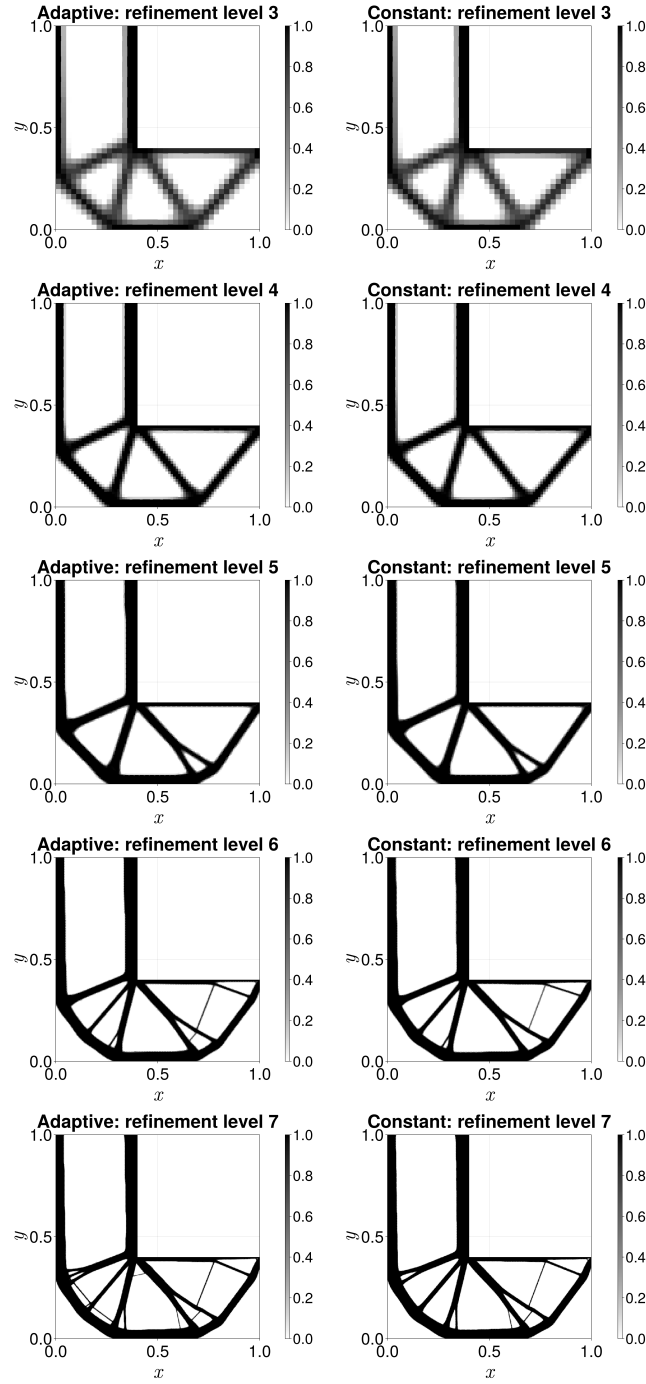


Figure 17: Comparison of the density field for the L-Cantilever for different refinement levels after 500 iterations

A.3 Cantilever

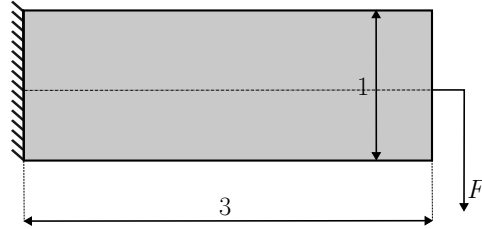


Figure 18: Geometry of the Cantilever

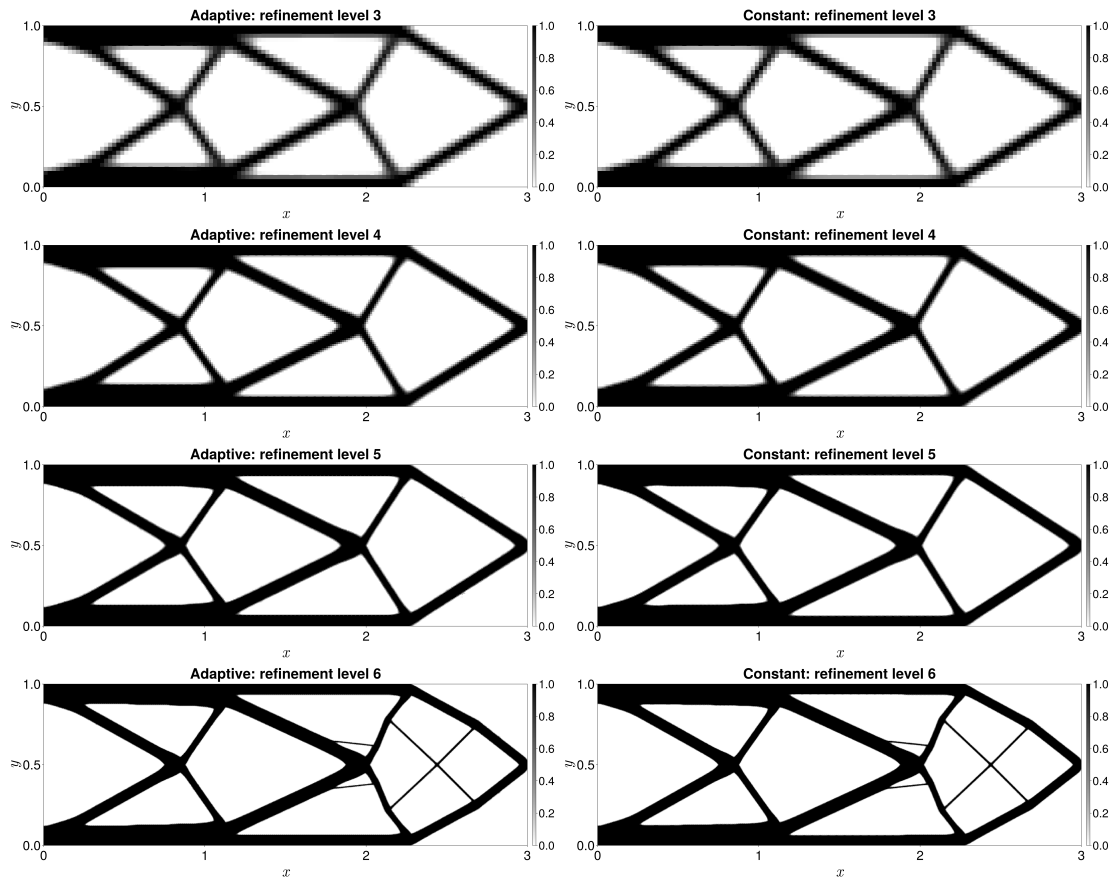


Figure 19: Comparison of the density field for the cantilever for different refinement levels after 500 iterations